

VGGT for 3D Reconstruction and Beyond: A Survey of Geometric State Strengthening and Its Applications

RUIYANG CHEN, Beijing University of Posts and Telecommunications, Beijing, China

FEIRAN LI, Independent Researcher

RUIYANG CHENG, Beijing University of Posts and Telecommunications, Beijing, China

JIASHUO YANG, Beijing University of Posts and Telecommunications, Beijing, China

CHU ZHOU, National Institute of Informatics, Tokyo, Japan

HENG GUO*, Beijing University of Posts and Telecommunications, Beijing, China

BOXIN SHI, Peking University, Beijing, China

ZHANYU MA, Beijing University of Posts and Telecommunications, Beijing, China

Visual Geometry Grounded Transformer (VGGT) jointly predicts cameras, point maps, depth maps, and tracks from multiple images in a feed-forward manner. *It provides multi-view geometry latent features and structured geometric outputs (i.e., cameras, point maps, depth maps, and tracks).* Denoted by $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$, these two components together form the geometric state examined in this survey. From this geometric-state perspective, we organize the literature into two directions: state-strengthening methods improve how this state is formed from diverse inputs, computed efficiently, made robust, retained over streams, or extended to dynamic scenes, whereas state-reuse methods use geometric state for novel-view synthesis, simultaneous localization and mapping (SLAM), semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Beyond the method taxonomy, we organize the datasets, metrics, and evaluation targets used to assess geometric-state strengthening and downstream reuse. Finally, we point out open challenges for future research. Our open-source resources are available at https://richardchen225.github.io/vggt_survey/.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Computer vision**; **Reconstruction**.

Additional Key Words and Phrases: 3D vision, geometric foundation models, feed-forward reconstruction, geometric state, point maps, visual geometry grounded transformer

1 Introduction

3D reconstruction is a long-standing problem in computer vision that provides the geometric foundation for downstream tasks such as novel-view synthesis [2, 75], localization and mapping [109, 110], semantic 3D scene understanding [131, 189], geometry-aware world models [129, 153], and embodied action and planning [1, 194]. Over the past several years, 3D reconstruction has progressed from scene-specific optimization with neural representations such as NeRF [115] and 3D Gaussian Splatting [80] toward feed-forward reconstruction represented by DUST3R-style [175] and VGGT-style [167] models.

Feed-forward reconstruction amortizes the reconstruction process across scenes, allowing a model to predict camera or scene geometry directly from sparse, unposed, or partially constrained observations rather than fitting a separate scene-specific model. A recent influential line of feed-forward reconstruction is represented by point-map-based models in the DUST3R family, including

*Corresponding author.

Authors' Contact Information: Ruiyang Chen, Beijing University of Posts and Telecommunications, Beijing, China, chenruiyang@bupt.edu.cn; Feiran Li, Independent Researcher, fr1994@hotmail.com; Ruiyang Cheng, Beijing University of Posts and Telecommunications, Beijing, China, eifw1001@gmail.com; Jiashuo Yang, Beijing University of Posts and Telecommunications, Beijing, China, heywhf@bupt.edu.cn; Chu Zhou, National Institute of Informatics, Tokyo, Japan, zhou_chu@hotmail.com; Heng Guo, Beijing University of Posts and Telecommunications, Beijing, China, guoheng@bupt.edu.cn; Boxin Shi, School of Computer Science, Peking University, Beijing, China, shiboxin@pku.edu.cn; Zhanyu Ma, Beijing University of Posts and Telecommunications, Beijing, China, mazhanyu@bupt.edu.cn.

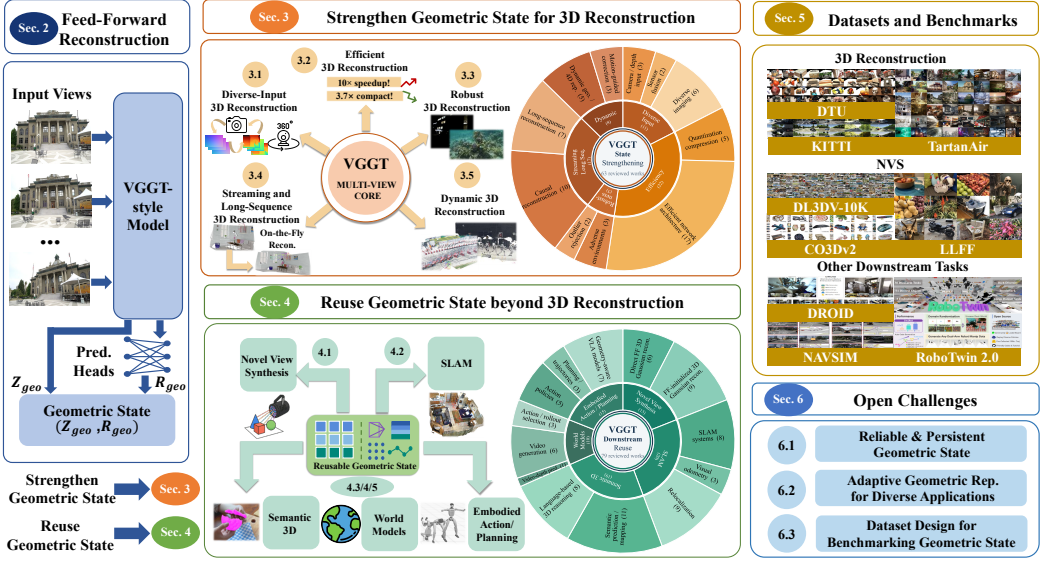


Fig. 1. Overview of the survey organization. Feed-forward reconstruction produces a geometric state containing multi-view geometry latent features Z_{geo} and structured geometric outputs R_{geo} . State-strengthening methods improve how this state is formed, computed, made robust, retained over streams, or extended to dynamic scenes. Downstream methods reuse geometric state for novel-view synthesis, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Datasets and evaluation targets assess the reviewed methods, and the open challenges concern reliable and persistent state, adaptive representations, and dataset design for benchmarking geometric state. Numbers in parentheses indicate the number of reviewed works in each category or subcategory.

DUST3R [175], MAST3R [89], Fast3R [188], MUST3R [12], and CUT3R [172], which made pixel-aligned 3D structure a practical interface between images and scene geometry. DUST3R [175] uses an image pair as its basic inference unit, so a larger image collection must still be assembled through global alignment. Building on the direct image-to-point-map formulation of DUST3R [175], the field has progressed to VGGT-style [167] models that jointly infer structured geometric outputs and learn reusable multi-view geometry latent features.

Although existing surveys have reviewed neural scene representations [182], 3D Gaussian Splatting [18], and feed-forward 3D reconstruction from DUST3R to VGGT [204, 208], the rapidly expanding VGGT [167] lineage—including numerous architectural extensions and downstream adaptations—has received limited systematic analysis. A dedicated survey is therefore needed to clarify how these works relate to one another, assess the current state of the field, and identify a roadmap for future research.

To organize this fragmented literature, we review VGGT-style [167] models through their geometric state, comprising multi-view geometry latent features Z_{geo} and structured geometric outputs R_{geo} . We divide existing studies into two branches: state strengthening and state reuse. As illustrated in Fig. 1, state strengthening improves how the geometric state is constructed and maintained, whereas state reuse adapts or consumes the state for downstream applications.

State strengthening. The first branch improves how the geometric state is formed from diverse inputs, computed efficiently, made robust, retained across long sequences, or extended to dynamic

scenes. Accordingly, Section 3 organizes these studies into diverse-input, efficient, robust, streaming and long-sequence, and dynamic 3D reconstruction.

State reuse. The second branch uses the structured geometric outputs, multi-view geometry latent features, or both components of the geometric state to support downstream applications beyond feed-forward reconstruction. Section 4 examines how 5 downstream application groups—novel-view synthesis, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning—use these components. Application models may use the state produced by VGGT [167] or by a strengthened variant, depending on the information required by the target application.

Datasets and benchmarks. To support meaningful comparison across both branches, Section 5 organizes the datasets, metrics, and evaluation targets used to assess geometric-state strengthening and downstream reuse. This analysis reveals substantial differences in evaluation coverage: static reconstruction and rendering have relatively mature resources, whereas long-stream, dynamic, semantic, embodied, and world-model settings depend more heavily on specialized datasets.

Open challenges. Based on the reviewed methods and evaluation evidence, Section 6 identifies 3 unresolved problems: building reliable and persistent geometric state, designing adaptive geometric representations for diverse applications, and designing datasets and benchmarks that determine whether application gains result from changes in or reuse of geometric state. Together, these challenges follow the survey’s progression from strengthening the geometric state, to reusing it in downstream applications, and finally to evaluating what the state contributes.

Contribution. To summarize, this survey provides the first systematic review centered on the VGGT [167] lineage, introduces a geometric-state perspective that unifies state strengthening and downstream reuse, consolidates the associated datasets and evaluation targets, and identifies key directions for future research. Our goal is to help researchers understand the rapidly evolving VGGT [167] landscape, relate existing methods within a common framework, and identify promising opportunities for further development in this growing field.

2 Feed-Forward 3D Reconstruction: Foundations and VGGT Model

This section first reviews the evolution from scene-specific optimization to feed-forward 3D reconstruction. It then introduces VGGT [167] and defines its geometric state. Finally, we discuss the model limitations that motivate Sections 3 and 4.

2.1 Evolution toward Feed-Forward 3D Reconstruction

Classical 3D reconstruction solves each scene through an optimization process. Structure-from-Motion (SfM) [139] estimates camera poses and sparse structure from image correspondences, after which Multi-View Stereo (MVS) [140] densifies the recovered geometry. NeRF [115] and 3D Gaussian Splatting (3DGS) [80] replace conventional geometric representations with neural radiance fields or optimizable Gaussian primitives, but their original formulations still fit scene-specific parameters from posed observations. Although these approaches differ in representation and rendering, geometry becomes available only after optimization has been performed for the target scene.

Feed-forward reconstruction moves reconstruction from per-scene optimization to amortized inference. After learning across many scenes, a model can predict camera or scene geometry directly from sparse, unposed, or partially constrained observations. The amortized formulation makes geometric structure an immediate network output rather than only the endpoint of a per-scene solver.

DUST3R [175] instantiated the feed-forward paradigm through point maps, assigning each image pixel a 3D coordinate in a common coordinate frame within an image pair. Its basic inference unit is

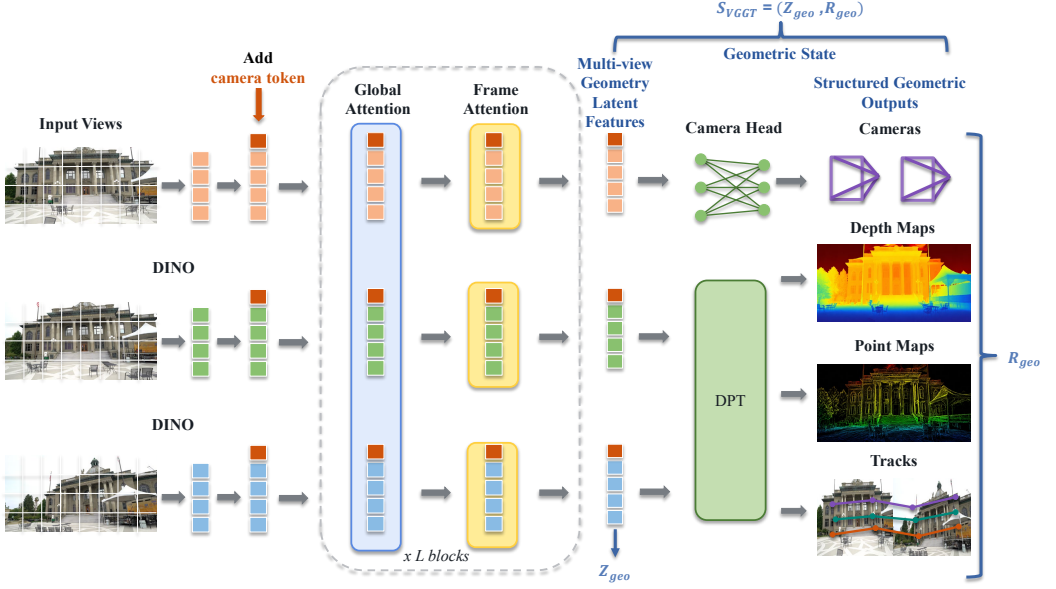


Fig. 2. VGGT [167] architecture. Input images are converted into tokens and augmented with camera and register tokens. Alternating frame-wise and global attention produces the multi-view geometry latent features Z_{geo} . Prediction heads decode cameras, depth maps, point maps, and tracks as the structured geometric outputs R_{geo} .

an image pair: the network predicts pixel-aligned point maps for both views, while reconstruction over a larger image collection requires predictions from multiple pairs to be assembled through a subsequent global alignment stage. Camera parameters can be recovered from the aligned geometry, but they are not jointly decoded with a complete multi-view scene representation in the same forward pass.

Point maps therefore establish a direct image-to-3D mapping, but DUST3R [175] still forms a coherent multi-view reconstruction by aligning pairwise predictions. This limitation motivates VGGT [167], which learns a set-level multi-view representation from which cameras, point maps, depth, and tracks can be decoded together. Its formulation and architecture are introduced next.

2.2 Introduction to VGGT

Given a set of input images, VGGT [167] aggregates multi-view evidence into a shared latent representation and predicts camera parameters, dense point maps, depth maps, and 3D point tracks in a single forward pass. This brings camera estimation, depth prediction, point-map reconstruction, and tracking into a single learned inference stage. Unlike pairwise point-map methods, it processes the image set jointly and exposes the resulting scene geometry through task-specific prediction heads.

VGGT [167] uses a multi-view transformer to aggregate information across the image set. Alternating frame-wise and global attention allows within-view structure and cross-view evidence to interact before camera, point-map, depth, and track heads make their predictions. Fig. 2 summarizes this architecture, from image tokenization and multi-view fusion to the 4 structured outputs.

More concretely, DINOv2 [118] patchifies each image into image tokens, to which VGGT [167] appends a camera token and register tokens for that view. The first view receives distinct learned

tokens so that it can serve as the reference frame. For an image set $\mathcal{I}_{\text{img}} = \{I_i\}_{i=1}^N$, the backbone can be summarized as

$$\begin{aligned} X_i^{(0)} &= [E_{\text{DINO}}(I_i), z_i^{\text{cam}}, Z_i^{\text{reg}}], \\ X^{(\ell+1)} &= \mathcal{A}_{\tau(\ell)}^{(\ell)}(X^{(\ell)}), \quad \tau(\ell) \in \{\text{frame}, \text{global}\}. \end{aligned} \quad (1)$$

Here I_i is the i th input image, N is the number of views, and E_{DINO} is the DINOv2 [118] image encoder that maps I_i to patch tokens. The learned token z_i^{cam} carries view-level camera information, while Z_i^{reg} denotes the bank of register tokens appended to view i ; $X_i^{(0)}$ is their concatenation with the patch tokens, and $X^{(\ell)} = \{X_i^{(\ell)}\}_{i=1}^N$ collects the tokens at transformer stage ℓ . The operator $\mathcal{A}_{\tau(\ell)}^{(\ell)}$ is the ℓ th attention block, and $\tau(\ell)$ specifies whether that block applies frame-wise or global attention. Frame-wise attention processes each $X_i^{(\ell)}$ separately, whereas global attention processes the tokens concatenated across views. The intermediate image tokens across these stages form the hierarchy later denoted by $\mathcal{Z}_{\text{geo}}$.

The prediction heads read different parts of this shared hierarchy. We denote the camera-head output for view i by $g_i = [q_i, t_i, f_i]$, where $q_i \in \mathbb{R}^4$ is a rotation quaternion, $t_i \in \mathbb{R}^3$ is translation, and $f_i = (f_i^h, f_i^v)$ contains the horizontal and vertical fields of view; the first camera defines the reference coordinate frame. A Dense Prediction Transformer (DPT) decoder [167] combines image tokens from multiple backbone stages to produce dense depth maps, point maps, tracking features, and confidence estimates for depth and points. The tracking module then consumes the dense tracking features together with a set of query points Q and predicts their cross-view locations and visibility. Thus tracks are query-conditioned outputs, whereas cameras, depth, and point maps are decoded directly for every input view.

The outputs are separate but architecturally coupled through end-to-end multi-task training, which can be schematically summarized as

$$\mathcal{L} = \mathcal{L}_{\text{camera}} + \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{pmap}} + \lambda_{\text{track}} \mathcal{L}_{\text{track}}. \quad (2)$$

Camera parameters are supervised with a robust regression loss; depth and point maps use confidence-weighted value and gradient terms; and the tracking objective includes correspondence and visibility supervision. The prediction heads share multi-view features through the backbone, but joint training does not by itself enforce exact geometric agreement among their predictions.

Taken together, the architecture exposes two related forms of geometric information. Before decoding, the intermediate transformer stages retain multi-view geometry latent features shaped by the joint camera, point-map, depth, and tracking objectives. After decoding, the prediction heads provide structured geometric outputs, including cameras, point maps, depth maps, and query-conditioned tracks, that can be inspected or used by another application model. This distinction is central to the later literature: methods may modify the internal representation, operate on the decoded geometry, or combine both components in a downstream application. The next subsection formalizes these two parts as the geometric state used throughout the survey.

2.3 Introduction to Geometric State

To compare methods that access different parts of VGGT [167], we formalize the *geometric state* as an analytical abstraction for this survey. This abstraction separates the internal multi-view representation from the structured quantities decoded from that representation:

$$\mathcal{S}_{\text{VGGT}} = (\mathcal{Z}_{\text{geo}}, \mathcal{R}_{\text{geo}}), \quad (3)$$

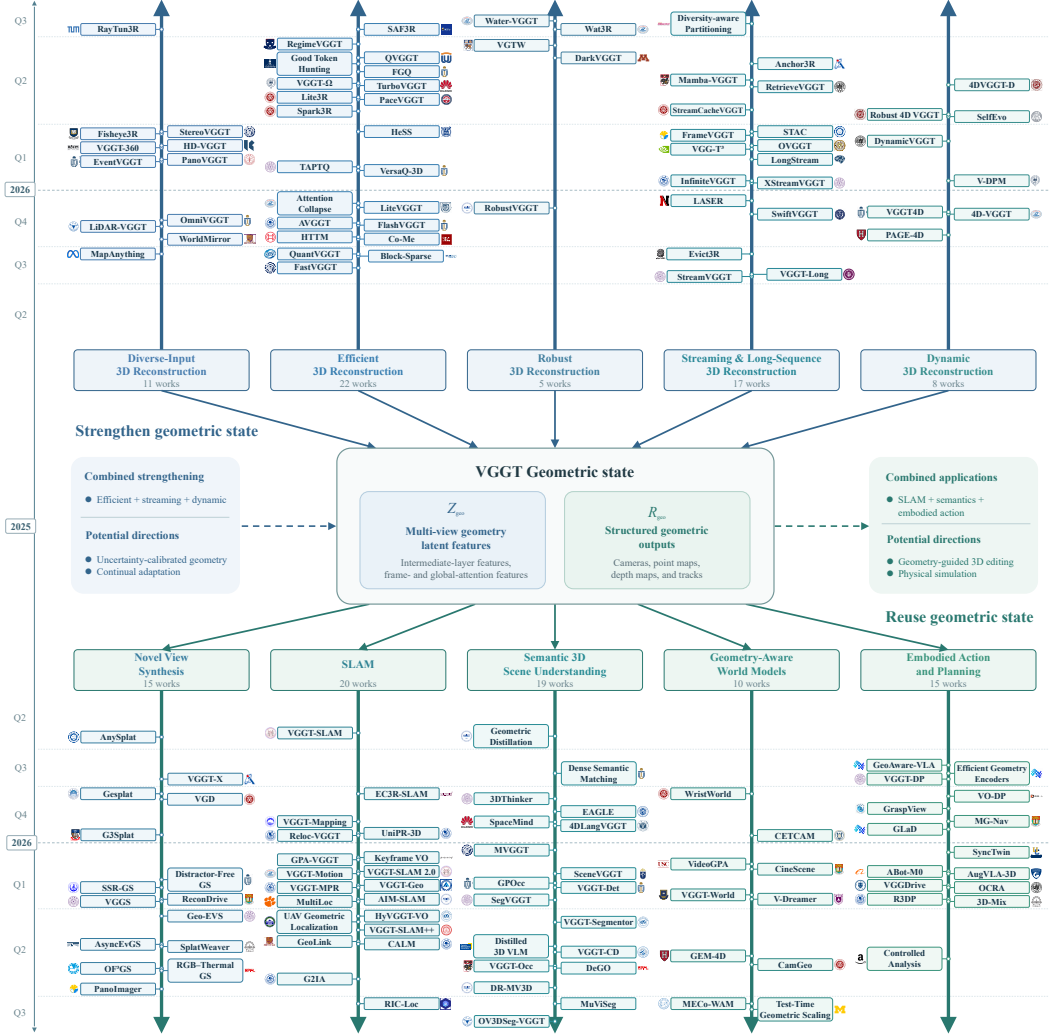


Fig. 3. Development of the VGGT [167] lineage around geometric state. The center contains multi-view geometry latent features Z_{geo} and structured geometric outputs R_{geo} . The 5 upper branches organize geometric-state strengthening methods, and the 5 lower branches organize downstream applications. Methods follow chronological order outward from the center, with years and quarters marked on the left. The side panels illustrate combinations of strengthening directions and applications, together with potential directions.

Equation (3) defines one geometric state with two components: Z_{geo} denotes multi-view geometry latent features, and R_{geo} contains structured geometric outputs. For VGGT [167], S_{VGGT} denotes the base geometric state. Later methods may strengthen how these components are formed, retained, or decoded.

The multi-view geometry latent features are collected from selected stages of the backbone:

$$Z_{geo} = \left\{ Z_{geo}^{(\ell)} \right\}_{\ell \in \mathcal{L}}, \quad (4)$$

where $\mathcal{L}$ denotes whichever backbone stages a particular method exposes or selects. Thus $\mathcal{Z}_{\text{geo}}$ denotes the selected feature levels rather than a fixed tensor shared by every method. These features are learned through multi-view aggregation and joint camera, point-map, depth, and track supervision; correspondence-like and epipolar structure observed in the attention layers provides additional evidence that these features encode more than appearance alone [8].

For N views and a set of track queries Q , $\mathcal{R}_{\text{geo}}$ collects the structured geometric outputs used by the later taxonomy:

$$\mathcal{R}_{\text{geo}}(Q) = \{C_i, P_i, D_i, T_i(Q)\}_{i=1}^N. \quad (5)$$

C_i denotes the camera pose together with intrinsics derived from the predicted fields of view; P_i is the image-aligned point map expressed in the first-camera reference frame; D_i is depth along the rays of view i under the model's shared normalized scale; and $T_i(Q)$ contains tracks conditioned on queries rather than an unconditional scene tensor. Visibility and the confidence maps associated with depth and points are auxiliary outputs and are left outside the compact state notation. Normals, Gaussian parameters, semantics, or other variant-specific predictions are treated as derived outputs rather than part of the base geometric state.

The two components support different uses of the geometric state. $\mathcal{Z}_{\text{geo}}$ carries cross-view information before it is committed to a particular decoded output, whereas $\mathcal{R}_{\text{geo}}$ exposes cameras, points, depth, and tracks in an explicit coordinate frame. When states are produced from separate windows, their structured geometric outputs may additionally require pose, scale, and orientation alignment; whether the state is frozen, fine-tuned, adapted, distilled, or jointly trained describes how it is transferred rather than which component is used.

Fig. 3 summarizes the development of the VGGT [167] lineage around geometric state. The upper branches organize the 5 strengthening groups in Section 3, and the lower branches organize the 5 application groups in Section 4. Within each branch, methods are arranged by when they appeared.

2.4 Limitations of VGGT

The limitations of VGGT [167] fall into two groups. The first concerns the conditions under which its geometric state can be formed and retained for reconstruction, motivating the strengthening directions in Section 3. The second concerns how that state can support objectives beyond reconstruction, motivating the downstream reuse studied in Section 4.

On the reconstruction side, the lack of a general mechanism for handling optional geometry, additional sensors, and non-standard imaging motivates diverse-input 3D reconstruction; growing per-forward-pass cost motivates efficient 3D reconstruction; failures under degraded or ambiguous visual observations motivate robust 3D reconstruction; finite context and inconsistent predictions across windows motivate streaming and long-sequence 3D reconstruction; and the static-scene formulation motivates dynamic 3D reconstruction. Each direction strengthens how $\mathcal{Z}_{\text{geo}}$ is formed or retained, how it produces $\mathcal{R}_{\text{geo}}$, or how both components evolve over time.

On the downstream side, even accurate reconstruction leaves a reconstruction-oriented representation rather than a complete solution for later applications. The structured geometric outputs do not directly provide appearance, persistent mapping, semantic identity, future transitions, or action grounding, while the multi-view geometry latent features must still be selected and adapted for these objectives. Section 4 therefore examines how application models support NVS, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Across these applications, the central issue shifts from improving state formation to selecting, adapting, or combining $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$ according to the target capability.

3 Strengthening Geometric State for 3D Reconstruction

Reconstruction methods strengthen one or both components of VGGT's [167] geometric state. They improve how $\mathcal{Z}_{\text{geo}}$ is formed and retained, how reliably it produces $\mathcal{R}_{\text{geo}}$, or how both components extend over time. We organize these advances into diverse-input, efficient, robust, streaming and long-sequence, and dynamic 3D reconstruction. Table 1 maps these categories to the affected state components, mechanisms, and outcomes.

3.1 Diverse-Input 3D Reconstruction

VGGT [167] is designed primarily for pinhole RGB images and does not directly accommodate known geometry, additional sensors, or non-standard imaging systems. We therefore organize diverse-input reconstruction into 3 directions: optional camera and depth as input incorporates cameras, depth, or partial geometry; geometry-aware sensor fusion combines visual features with stereo or LiDAR measurements; and diverse imaging systems adapt reconstruction to high-resolution, fisheye, panoramic, or event inputs.

Optional camera and depth as input. Methods in this group differ in how optional camera, depth, or partial-geometry inputs are represented before multi-view fusion. OmniVGGT [127] injects encoded camera parameters and depth tokens, with stochastic training supporting different input combinations. MapAnything [78] factorizes calibration, poses, depth, and partial reconstructions into ray directions, ray depths, camera poses, and metric scale for a common VGGT-style [167] transformer. WorldMirror [104] jointly tokenizes intrinsics, poses, depth, and image features, then predicts several geometric and rendering outputs through task-specific heads. The key design choice is therefore whether optional geometry enters as additional tokens, factored geometric quantities, or a unified multi-task input representation.

Geometry-aware sensor fusion. StereoVGGT [24] and LiDAR-VGGT [171] extend the input setting beyond standard monocular RGB collections. Frozen VGGT [167] features provide useful camera-calibration knowledge for stereo, but direct use can lose the fine spatial detail needed for binocular matching. StereoVGGT [24] therefore applies a training-free feature-adjustment pipeline to restore spatial detail while exploiting the calibration information encoded in the frozen features. LiDAR-VGGT [171] couples LiDAR-inertial odometry with VGGT [167] through coarse-to-fine fusion, using metric sensing to constrain scale and large-environment consistency while retaining dense visual geometry.

Diverse imaging systems. Non-standard images require changes at different stages of reconstruction. HD-VGGT [21] separates coarse global inference from reliability-aware refinement to preserve high-resolution detail. For fisheye cameras, Fisheye3R [42] adapts reconstruction directly, whereas RayTun3R [147] corrects positional encodings biased toward pinhole cameras. Panoramic methods either reconstruct perspective slices and reproject them, as in VGGT-360 [197], or learn spherical encoding and rotation-aware training directly, as in PanoVGGT [60]. EventVGGT [135] changes the feature extraction stage for event-based depth. The required modification therefore depends on whether the input differs in resolution, camera projection, or sensing process.

The 3 groups broaden reconstruction through optional geometric information, additional sensor measurements, and diverse image formation, respectively. All expand the evidence available to form $\mathcal{Z}_{\text{geo}}$ and predict $\mathcal{R}_{\text{geo}}$, but none guarantees that every supplied measurement is preserved exactly in the final structured geometric outputs.

3.2 Efficient 3D Reconstruction

Dense multi-view processing in VGGT [167] requires substantial computation and memory as the number or resolution of input views grows. We therefore organize efficient reconstruction into

Table 1. Strengthening geometric state in 3D reconstruction. Different columns show what part of the state is strengthened, representative methods, how the strengthening is performed, and the resulting capability.

What is strengthened	Representative works	How it is strengthened	Result
Sec. 3.1 Diverse-Input 3D Reconstruction			
Z_{geo}: input evidence from optional geometry and additional sensors $Z_{\text{geo}}, \mathcal{R}_{\text{geo}}$: camera models and image-formation settings	(i) Optional camera and depth as input: OmniVGGT [127] / MapAnything [78] / WorldMirror [104] (ii) Geometry-aware sensor fusion: StereoVGGT [24] / LiDAR-VGGT [171] (iii) Diverse imaging systems: HD-VGGT [21] / Fisheye3R [42] / VGGT-360 [197] / PanoVGGT [60] / EventVGGT [135] / RayTun3R [147]	(i) Encode known cameras, depth, poses, or partial geometry (ii) Fuse stereo or LiDAR measurements with visual features (iii) Adapt feature extraction and geometric decoding to resolution, projection, or sensing modality	- Reconstruct from more input types - Use metric or partial-geometry constraints - Support non-standard cameras and imaging systems
Sec. 3.2 Efficient 3D Reconstruction			
Z_{geo}: token and feature computation $Z_{\text{geo}} \rightarrow \mathcal{R}_{\text{geo}}$: geometric information retained after compression	(i) Quantization-based compression: QuantVGGT [46] / VersaQ-3D [210] / TAPTQ [121] / FGQ [209] / QVGGT [124] (ii) Efficient network architecture design: FastVGGT [143] / LiteVGGT [145] / HTTM [176] / FlashVGGT [178] / Block-Sparse [165] / Co-Me [23] / AVGGT [154] / Attention Collapse [90] / HeSS [82] / VGGT-Ω [168] / PaceVGGT [91] / Spark3R [157] / Good Token Hunting [215] / RegimeVGGT [193] / TurboVGGT [69] / SAF3R [35] / Lite3R [203]	(i) Quantize weights and activations with sensitivity-aware precision (ii) Merge, prune, or select tokens and use sparse or compressed attention and compact heads	- Reduce runtime and memory per forward pass - Retain camera, depth, and reconstruction accuracy after compression
Sec. 3.3 Robust 3D Reconstruction			
Z_{geo}: evidence formed from degraded or contaminated images $Z_{\text{geo}} \rightarrow \mathcal{R}_{\text{geo}}$: reliability of predicted geometry	(i) Adverse environments: DarkVGGT [84] / Water-VGGT [105] / Wat3R [134] (ii) Distractor and outlier rejection: VGTW [122] / RobustVGGT [61]	(i) Add thermal observations, physical constraints, or domain adaptation (ii) Suppress distractors and reject irrelevant or non-overlapping views	- Recover more reliable geometry in darkness or underwater scenes - Prevent distractors and outlier views from corrupting reconstruction
Sec. 3.4 Streaming and Long-Sequence 3D Reconstruction			
Z_{geo}: historical information retained between windows $\mathcal{R}_{\text{geo}}$: scale and coordinate consistency across windows	(i) Causal reconstruction: StreamVGGT [224] / FrameVGGT [186] / STAC [173] / InfiniteVGGT [198] / OVGGT [107] / Evict3R [111] / XStreamVGGT [150] / StreamCacheVGGT [101] / RetrieveVGGT [225] / Mamba-VGGT [39] (ii) Long-sequence reconstruction: LASER [40] / LongStream [26] / Anchor3R [158] / VGGT-Long [36] / SwiftVGGT [87] / Diversity-aware Partitioning [125] / VGG-T³ [43]	(i) Select, compress, retrieve, or propagate history during causal updates (ii) Divide long inputs into local windows and align their scale and coordinates using anchors, loop closure, or specialized training	- Process causal streams and image collections larger than a single inference window - Limit memory while maintaining globally aligned cameras and geometry
Sec. 3.5 Dynamic 3D Reconstruction			
Z_{geo}: features that separate persistent structure from scene change $\mathcal{R}_{\text{geo}}$: time-varying geometry and motion	(i) Dynamic geometry and 4D representation: PAGE-4D [220] / 4D-VGGT [166] / V-DPM [151] / SelfEvo [71] / DynamicVGGT [63] (ii) Motion-guided geometry correction: VGGT4D [68] / Robust 4D VGGT [200] / 4DVGGT-D [201]	(i) Fuse information across space and time and predict dynamic or future geometry (ii) Detect moving regions and reduce their effect on camera, depth, and point predictions	- Stabilize camera and geometry estimates in dynamic scenes - Represent scene change and, where modeled, predict dense 3D motion

two directions: quantization-based compression reduces the numerical cost of model weights and activations, while efficient network architecture design reduces redundant token processing and attention computation.

Quantization-based compression. The central question is where reduced precision can be applied without damaging geometric prediction. QuantVGGT [46] and VersaQ-3D [210] identify activation outliers, unstable calibration, and heterogeneous precision requirements as the main obstacles to generic quantization. TAPTQ [121] compensates modules with long-tailed activation error, FGQ [209] allocates precision using task-, block-, and channel-level Fisher sensitivity, and QVGGT [124] combines block-wise mixed precision with special-token filtering and consistency-aware scale search. The progression is from uniform compression toward sensitivity-aware precision allocation around the pathways responsible for camera, depth, and point predictions.

Efficient network architecture design. Beyond quantization-based compression, a second line reduces architectural redundancy. FastVGGT [143], LiteVGGT [145], Co-Me [23], and PaceVGGT [91] merge or prune tokens using similarity, geometric confidence, or learned pre-attention importance. Spark3R [157], Good Token Hunting [215], and RegimeVGGT [193] distinguish query from key-value roles, frame from patch selection, or shallow, middle, and deep layer regimes. HTTM [176], FlashVGGT [178], block-sparse global attention [165], HeSS [82], TurboVGGT [69], and SAF3R [35] instead restructure or adapt attention across heads, layers, and inputs; Lite3R [203] combines sparse linear attention with low-precision training. Analyses by AVGGT [154] and the attention-collapse study [90] explain why such reductions must remain selective, while VGGT- Ω [168] uses a unified dense head and scene tokens for feature aggregation. The shared logic is to reduce token processing and attention computation across heads, layers, and views while retaining the cross-view geometric information needed by the prediction heads.

The two directions reduce the cost of a forward pass in different ways: quantization lowers numerical cost, whereas efficient network architecture design reduces token processing and attention computation. Both aim to form $\mathcal{Z}_{\text{geo}}$ more efficiently without losing the cross-view information needed for stable $\mathcal{R}_{\text{geo}}$.

3.3 Robust 3D Reconstruction

VGGT [167] can become unreliable when visual evidence is degraded by the environment or contaminated by distractors and inconsistent views. We therefore organize robust reconstruction into two directions: methods for adverse environments recover or supplement weak visual evidence, while distractor and outlier rejection prevents irrelevant content or views from corrupting multi-view fusion.

Adverse environments. The appropriate robustness mechanism depends on why visual evidence is degraded. DarkVGGT [84] supplements missing visible-light structure with RGB-thermal observations. Underwater reconstruction instead addresses attenuation and scattering: WaterVGGT [105] imposes water-optics constraints, while Wat3R [134] adapts geometry from unlabeled underwater video through cross-view consistency. Thermal sensing provides additional observations in darkness, whereas physical constraints and adaptation address systematic environmental distortion.

Distractor and outlier rejection. VGTW [122] suppresses transient distractors and occlusions during cross-view fusion. RobustVGGT [61] addresses irrelevant or non-overlapping input views by exploiting emergent outlier suppression in VGGT [167] attention and intermediate features. It filters inconsistent views with a global threshold and reruns reconstruction without fine-tuning. Both methods improve robustness to contaminated image sets by identifying inputs that should contribute less to $\mathcal{Z}_{\text{geo}}$.

The two groups address different causes of unreliable input evidence: adverse environments degrade otherwise relevant observations, whereas input contamination introduces views or content that should be rejected. Existing methods improve the reliability of $\mathcal{R}_{\text{geo}}$ by supplementing weak evidence or suppressing inconsistent evidence. Coverage remains condition-specific: darkness,

underwater scenes, and distractors have dedicated solutions, whereas rain, fog, and compound weather corruption remain less studied within the current VGGT [167] literature.

3.4 Streaming and Long-Sequence 3D Reconstruction

VGGT [167] jointly processes a bounded image set and therefore cannot update its reconstruction causally as new frames arrive or maintain a globally consistent reconstruction over long sequences. We therefore organize this subsection into two directions: causal reconstruction updates geometry online as each frame arrives, while long-sequence reconstruction connects bounded local predictions into a consistent global result.

Causal reconstruction. The first direction addresses online reconstruction with an unknown sequence length: the model must update the current geometry before seeing future observations. StreamVGGT [224] introduces causal attention with a historical key-value (KV) cache, FrameVGGT [186] organizes bounded context into frame-wise units, and STAC [173] compresses past information using temporal salience and voxel-aligned spatial aggregation. InfiniteVGGT [198], OVGGT [107], and Evict3R [111] bound the retained context through rolling selection, anchor protection, or training-free eviction. XStreamVGGT [150] further combines KV pruning and quantization, while StreamCacheVGGT [101] selectively merges cached tokens to reduce cache size. RetrieveVGGT [225] rebuilds a fixed context through retrieval, and Mamba-VGGT [39] propagates a compact external state. These designs strengthen $\mathcal{Z}_{\text{geo}}$ by selecting, compressing, retrieving, or propagating the past evidence most useful for each causal update.

Long-sequence reconstruction. The second direction addresses long inputs whose observations cannot be processed in one bounded forward pass. Individually accurate local predictions can become mutually inconsistent after many state updates, so long-sequence methods must both partition observations and reconnect their outputs. LASER [40] aligns layer-dependent scales across adjacent windows, while LongStream [26] learns keyframe-relative poses, orthogonal scale, cache consistency, and periodic refresh. Anchor3R [158] avoids a persistent first-frame gauge: it predicts current-centric local measurements and uses transient anchors, motion averaging, and loop-closure reinsertion to recover a global map. VGGT-Long [36] combines overlapping chunks with lightweight loop closure, while SwiftVGGT [87] detects loops from backbone features and applies single-step Sim(3) alignment. Diversity-aware View Partitioning [125] balances visual dissimilarity and spatial dispersion when constructing chunks. VGG-T³ [43] distills variable-length KV geometry into a fixed-size multilayer perceptron (MLP) state through test-time training. These methods strengthen $\mathcal{R}_{\text{geo}}$ by keeping decoded cameras and geometry in a consistent scale and coordinate system as the sequence grows.

Together, the two directions address online state updates and globally consistent reconstruction over inputs that exceed a bounded inference window. Causal reconstruction retains the evidence needed for each update, while long-sequence reconstruction aligns the resulting local predictions across the full sequence.

3.5 Dynamic 3D Reconstruction

VGGT [167] assumes a static scene, so moving content can corrupt camera and geometry estimates and cannot be represented as time-varying structure. We therefore organize dynamic 3D reconstruction into two directions: the dynamic geometry and 4D representation direction explicitly models scene structure and motion over time, while motion-guided geometry correction identifies dynamic or uncertain regions to stabilize pose and geometry.

Dynamic geometry and 4D representation. The first direction represents scene changes within the geometric state rather than only correcting their effect on static reconstruction. PAGE-4D [220] uses dynamics-aware masks to route motion evidence differently for pose and geometry,

and 4D-VGGT [166] factorizes cross-view spatial fusion and cross-time local fusion. V-DPM [151] makes the extension explicit with time-varying dynamic point maps and dense 3D motion, together with camera and geometry prediction in the full model. SelfEvo [71] addresses limited 4D supervision: its spatio-temporal self-distillation adapts pretrained reconstruction models, including VGGT [167], using unlabeled videos rather than scarce 4D annotations. Driving scenes provide a demanding instance of this technical direction because they contain many moving objects and require temporally consistent geometry over long trajectories. DynamicVGGT [63] addresses this setting by jointly predicting current and future point maps, applying motion-aware temporal attention, and modeling point motion with a dynamic Gaussian head.

Motion-guided geometry correction. The second direction retains the pretrained VGGT representation but reduces the effect of moving regions on VGGT [167] predictions. VGGT4D [68] mines layer-wise motion cues from global attention and feeds refined dynamic masks back into early inference. Robust 4D VGGT [200] combines attention-derived uncertainty with local geometry purification, whereas 4DVGGT-D [201] first stabilizes pose against dynamic interference and then refines static and dynamic depth. These training-free methods detect motion or uncertainty and use it to correct pose, depth, or point-map predictions. The two directions respectively represent scene change within the geometric state and correct base geometry using motion cues.

Dynamic variants make both forms of geometric information time dependent: the multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$ must distinguish persistent structure from change, while the structured geometric outputs $\mathcal{R}_{\text{geo}}$ expose time-varying geometry and, where modeled, motion without destabilizing the coordinate frame.

3.6 Summary of Geometric State Strengthening

This section has shown how geometric state is strengthened through broader inputs, lower computation, greater robustness, retained streaming context, and explicit scene dynamics. These directions are not independent: streaming and long-sequence reconstruction depend on efficient computation, while online reconstruction of dynamic scenes also requires memory mechanisms that preserve context without treating motion as static structure. Combining multiple strengthening directions therefore broadens both the conditions under which $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$ remain useful and the information that later applications can obtain from the state. Section 4 next examines those applications.

4 Reusing Geometric State Beyond 3D Reconstruction

Downstream applications reuse the structured geometric outputs $\mathcal{R}_{\text{geo}}$, the multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$, or their combination. We organize this literature into 5 applications: NVS, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Table 2 summarizes the geometric-state information used in each application and the corresponding method groups.

4.1 Novel View Synthesis

Novel-view synthesis converts the geometric state into a renderable scene representation for synthesizing unseen views. With 3D Gaussian Splatting [80] as the dominant representation, methods differ in whether feed-forward geometry completes the reconstruction or provides a starting point for later refinement. We therefore organize them into direct feed-forward 3D Gaussian reconstruction and 3D Gaussian reconstruction initialized by feed-forward geometry.

Direct feed-forward 3D Gaussian reconstruction. This route decodes the Gaussian scene without first optimizing a separate representation from structured geometric outputs. AnySplat [75], VGD [96], and SplatWeaver [163] train feed-forward Gaussian heads with VGGT [167]-derived

Table 2. Geometric-state reuse across 5 downstream applications: novel view synthesis (NVS), SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Different columns show which geometric-state components and information are used in each application and list the corresponding method groups.

Geometric-state components and information used	Method groups and representative works
Sec. 4.1 Novel View Synthesis	
$\mathcal{R}_{\text{geo}}$: cameras, depth, and point maps used to construct or initialize renderable geometry $\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features decoded into appearance or Gaussian parameters	Direct feed-forward 3D Gaussian reconstruction: AnySplat [75] / VGD [96] / SplatWeaver [163] / G3Splat [66] / ReconDrive [195] / OF³GS [19] 3D Gaussian reconstruction initialized by feed-forward geometry: VGGT-X [103] / Gesplat [106] / VGGs [181] / Geo-EVS [85] / Panolmager [187] / SSR-GS [44] / Distractor-Free GS [56] / AsyncEvGS [31] / RGB-Thermal GS [28]
Sec. 4.2 SLAM	
$\mathcal{R}_{\text{geo}}$: cameras, depth, and point maps used for tracking, alignment, and mapping $\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features used for correspondence, retrieval, and place recognition	SLAM systems: VGGT-SLAM [110] / VGGT-SLAM 2.0 [109] / VGGT-SLAM++ [112] / EC3R-SLAM [67] / AIM-SLAM [74] / VGGT-Geo [130] / VGGT-Mapping [156] / CALM [206] Visual odometry: VGGT-Motion [183] / HyVGGT-VO [120] / Keyframe VO [32] Relocalization: VGGT-MPR [184] / UniPR-3D [37] / GPA-VGGT [185] / Reloc-VGGT [38] / MultiLoc [33] / RIC-Loc [77] / G2IA [76] / UAV Geometric Localization [92] / GeoLink [202]
Sec. 4.3 Semantic 3D Scene Understanding	
$\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features used for segmentation, language reasoning, and association $\mathcal{R}_{\text{geo}}$: 3D locations and geometry used to ground semantic predictions	3D semantic prediction and mapping: SegVGGT [131] / VGGT-Segmentor [49] / VGGT-Det [15] / OV3DSeg-VGGT [218] / VGGT-Occ [22] / GPOcc [217] / DeGO [48] / Dense Semantic Matching [189] / MuViSeg [45] / SceneVGGT [53] / VGGT-CD [207] Language-based 3D scene reasoning: MVGGT [179] / 4DLangVGGT [180] / SpaceMind [212] / 3DThinker [25] / Geometric Distillation [88] / Distilled 3D VLM [4] / DR-MV3D [27] / EAGLE [14]
Sec. 4.4 Geometry-Aware World Models	
$\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features predicted over time or used to condition and supervise generation $\mathcal{R}_{\text{geo}}$: cameras, depth, and point maps used to condition or evaluate future predictions	Geometry-conditioned video-depth prediction: VGGT-World [153] Geometry-conditioned video generation: CETCAM [214] / CineScene [70] / CamGeo [100] / WristWorld [129] / VideoGPA [41] / V-Dreamer [62] Geometry-guided action prediction and rollout selection: MECo-WAM [205] / GEM-4D [219] / Test-Time Geometric Scaling [213]
Sec. 4.5 Embodied Action and Planning	
$\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features fused into policies or geometry-aware VLA models $\mathcal{R}_{\text{geo}}$: depth, points, or camera geometry used for planning and feasible action	Geometry-enhanced action policies: VGGT-DP [50] / R3DP [211] / VO-DP [117] / Efficient Geometry Encoders [162] / OCRA [170] Geometry-based planning and trajectory generation: GraspView [174] / SyncTwin [72] / MG-Nav [164] Geometry-aware VLA models: 3D-Mix [194] / GeoAware-VLA [1] / GLaD [59] / AugVLA-3D [132] / ABot-M0 [191] / VGGDrive [169] / Controlled Analysis [190]

initialization or distillation; SplatWeaver further distills camera and depth supervision while allocating primitives adaptively. G3Splat [66] regularizes the geometry of the decoded Gaussians, ReconDrive [195] adds spatio-temporal heads for dynamic driving scenes, and OF³GS [19] incrementally predicts, refines, and fuses Gaussians from a frozen StreamVGGT [224] front end. These variants trade a separate scene-optimization stage for direct or incremental inference.

3D Gaussian reconstruction initialized by feed-forward geometry. This route retains a subsequent reconstruction or optimization stage after cameras, points, or depth have been inferred. VGGT-X [103] scales and globally aligns VGGT [167] outputs before robust 3DGS [80] training; Gesplat [106] and VGGs [181] initialize pose-free or sparse-view reconstruction from cameras, points, or depth. The same principle extends to extrapolative driving views in Geo-EVS [85], sparse panoramic views in Panolmager [187], glossy surfaces in SSR-GS [44], and transient-content removal in Distractor-Free GS [56]. AsyncEvGS [31] and unpaired RGB-thermal GS [28] additionally use the inferred geometry to align event or thermal observations. Feed-forward structured geometric

outputs provide the geometric starting point, while the retained reconstruction stage refines appearance and adapts to the target setting.

The distinction is whether Gaussians are reconstructed directly from the feed-forward representation or refined after initialization from $\mathcal{R}_{\text{geo}}$. Both routes must preserve cross-view structure when appearance, dynamics, or sensing modality changes.

4.2 SLAM

SLAM uses VGGT [167] or its long-sequence extensions as geometric front ends to build consistent maps and camera trajectories. Unlike the reconstruction methods reviewed in Section 3.4, SLAM adds the system components needed for joint mapping and tracking, camera-motion estimation without full mapping, and pose recovery in previously observed scenes. We therefore organize these methods into SLAM systems, visual odometry, and relocalization.

SLAM systems. The first group uses VGGT [167] as a dense front-end and adds the tracking, submap alignment, and global optimization needed to construct a consistent map. VGGT-SLAM [110], VGGT-SLAM 2.0 [109], and VGGT-SLAM++ [112] align feed-forward submaps with projective or global optimization. EC3R-SLAM [67] and AIM-SLAM [74] reduce redundant inference through sparse tracking or informative keyframes, while VGGT-Geo [130] adds confidence-aware geometric constraints and VGGT-Mapping [156] supports robot navigation. CALM [206] extends this role to kilometer-scale mapping by using an assistant camera baseline, intrinsic correction, and anchor propagation to resolve scale and nonlinear submap distortion. Across these systems, $\mathcal{R}_{\text{geo}}$ is useful only after local gauges and accumulated drift are reconciled.

Visual odometry. Unlike SLAM systems, these methods focus on estimating camera motion without constructing and globally optimizing a persistent map. VGGT-Motion [183] and HyVGGT-VO [120] couple dense geometry to long-range camera tracking or visual odometry, while keyframe-based feed-forward VO [32] learns which frames to include in the VGGT [167] window. Per-window pose accuracy alone is insufficient: temporal selection, scale alignment, and global consistency determine whether small local errors accumulate along the estimated trajectory.

Relocalization. Place-recognition methods first identify relevant references: VGGT-MPR [184] uses VGGT [167] embeddings, depth-assisted LiDAR densification, and tracking-based re-ranking; UniPR-3D [37] aggregates 2D and 3D VGGT [167] tokens for multi-view retrieval; G2IA [76] combines VGGT [167] geometry priors with instance-aware candidate verification; and GeoLink [202] distills offline VGGT [167] geometry into a 2D-only cross-view retrieval representation. Pose relocalization is addressed by GPA-VGGT [185], Reloc-VGGT [38], and MultiLoc [33] through VGGT [167] adaptation or multi-reference pose estimation, while RIC-Loc [77] derives multiple posed-reference hypotheses and uses their consensus and dispersion for selective localization. UAV geometric localization [92] reconstructs local 3D structure and renders a bird's-eye-view intermediary for cross-view 3-DoF pose regression. This route is hybrid: $\mathcal{Z}_{\text{geo}}$ narrows or transfers the search representation, while $\mathcal{R}_{\text{geo}}$ establishes geometric compatibility.

SLAM therefore adds 3 capabilities to feed-forward or streaming geometry: globally consistent mapping, visual odometry, and recovery of pose in a previously observed scene. Structured geometric outputs anchor geometry and pose, while multi-view geometry latent features support correspondence and retrieval.

4.3 Semantic 3D Scene Understanding

Semantic 3D scene understanding changes the target from reconstructing structure to attaching categories, language, and persistent identities to it. Semantic heads read multi-view spatial features from $\mathcal{Z}_{\text{geo}}$, while $\mathcal{R}_{\text{geo}}$ anchors their predictions in 3D. We therefore organize these methods into two directions: 3D semantic prediction and mapping; and language-based 3D scene reasoning.

3D semantic prediction and mapping. This group predicts semantic labels for objects, visible regions, or occupied 3D space and maintains those predictions across views or time. SegVGGT [131], VGGT-Segmentor [49], and VGGT-Det [15] address instance segmentation, cross-view segmentation, or multi-view detection. OV3DSeg-VGGT [218] extends segmentation to open-vocabulary categories. Beyond visible surfaces, VGGT-Occ [22] decodes semantic occupancy from a frozen multi-view encoder, GPOcc [217] converts surface points and features into sparse Gaussian occupancy, and DeGO [48] distills cross-view and cross-time features into deformable Gaussians. Persistent semantic mapping extends these outputs across views or time: Dense Semantic Matching [189] associates pixels across object instances, MuViSeg [45] matches whole segments jointly across multiple views, SceneVGGT [53] retains instance identities and timestamps in an online 3D map, and VGGT-CD [207] registers observations from different epochs to identify physical scene changes.

Language-based 3D scene reasoning. Methods differ in whether geometry enters through explicit grounding, feature fusion, or distillation. MVGGT [179] and 4DLangVGGT [180] ground referring expressions across views or time. SpaceMind [212] fuses VGGT [167] features under camera guidance, whereas 3DThinker [25] aligns a reasoning latent with VGGT [167] geometry. Geometric Distillation [88] transfers correspondence, relative-depth, and cost-volume cues into a vision-language model (VLM), and Distilled 3D VLM [4] adopts VGGT [167] as the encoder of a compact reasoning model. DR-MV3D [27] supervises global-map construction and view planning with geometry-consistent pseudo targets. EAGLE [14] localizes visual queries in 3D by combining episodic appearance and geometry memories. Each mechanism connects language or query outputs to cross-view spatial relations, but at a different stage of the reasoning model.

Together, the two directions progress from assigning meaning to 3D structure to reasoning about that structure with language or query memories. They typically combine transferable multi-view geometry latent features in $\mathcal{Z}_{\text{geo}}$ with the spatial anchors provided by structured geometric outputs in $\mathcal{R}_{\text{geo}}$.

4.4 Geometry-Aware World Models

Geometry-aware world models differ in what they predict and where geometric state enters the prediction process. Geometry can be predicted as future depth, guide the generation of future visual observations, or support action prediction and the selection of geometrically consistent rollouts. We therefore organize these methods into geometry-conditioned video-depth prediction, geometry-conditioned video generation, and geometry-guided action prediction and rollout selection.

Geometry-conditioned video-depth prediction. VGGT-World [153] takes RGB observations as input and predicts future video depth by evolving $\mathcal{Z}_{\text{geo}}$ autoregressively rather than predicting future pixels alone. The model uses the current multi-view geometry latent features to forecast a future geometric state, and structured geometric outputs then decode or verify the depth and camera geometry implied by the predicted state. This setting directly tests whether geometric state can support temporally coherent depth prediction beyond the observed frames.

Geometry-conditioned video generation. Geometry can condition generation as explicit tokens, multi-view geometry latent features, distilled supervision, or an evaluation signal. CET-CAM [214] tokenizes predicted cameras and depth, whereas CineScene [70] injects multi-view geometry latent features. CamGeo [100] distills camera, depth, and cross-frame constraints during training. WristWorld [129] reconstructs poses and 4D point maps before generating future wrist-camera views, while VideoGPA [41] forms preference pairs from a VGGT [167] reconstruction score. V-Dreamer [62] uses video generation priors to synthesize robot interaction videos and then converts their motion into kinematically aligned trajectories. Although the resulting trajectories support planning, its core predictive mechanism is video generation, so we classify it as a world

model. The relevant evidence is whether each form of generation preserves coherent structure under camera, scene, or robot motion and yields useful downstream trajectories where applicable.

Geometry-guided action prediction and rollout selection. MECo-WAM [205] transfers relational 4D targets from a frozen VGGT [167] encoder into a video-action world model. GEM-4D [219] distills dense correspondence from a pretrained 4D geometry model into video prediction and converts consistent rollouts into robot trajectories; VGGT [167] is evaluated as a teacher choice rather than assumed to be its default geometry model. Test-Time Geometric Scaling [213] uses frozen VGGT- Ω [168] camera, depth, and point-map predictions to rank sampled world-action rollouts by reprojection consistency and selectively spend additional computation. These methods use geometry to supervise action prediction or select a geometrically consistent rollout; their value must ultimately appear in both geometric consistency across predicted frames and improved action success.

Visual fidelity alone is insufficient for these methods. Future observations must preserve 3D structure, and action-conditioned models must show that this structure improves an action prediction or reliably identifies a better rollout.

4.5 Embodied Action and Planning

Embodied methods use geometric state to choose, plan, or execute actions that affect an environment. Different action models incorporate geometry at different stages, from policy features to explicit motion planning and multimodal action reasoning. We therefore organize them into geometry-enhanced action policies, geometry-based planning and trajectory generation, and geometry-aware VLA models.

Geometry-enhanced action policies. The design choice is how geometric evidence enters action prediction. VGGT-DP [50] and R3DP [211] fuse multi-view geometry latent features with the policy, VO-DP [117] balances semantic and geometric observations, and Efficient Geometry Encoders [162] reduce the latency of this transfer. OCRA [170] adds object-centric point clouds and tactile priors instead of relying on dense features alone. In these policies, $\mathcal{Z}_{\text{geo}}$ provides multi-view spatial features and selected structured geometric outputs provide explicit geometry for contact-related action prediction.

Geometry-based planning and trajectory generation. These methods use reconstructed geometry directly to select views, plan motion, or generate feasible trajectories. GraspView [174] uses reconstruction for view selection and grasp alignment, SyncTwin [72] maintains a registered digital twin, and MG-Nav [164] aligns observations and goals in spatial memory. Structured geometric outputs, alone or together with multi-view geometry latent features, connect observations and memory to feasible motion in these systems.

Geometry-aware VLA models. These methods add geometric information to vision-language-action (VLA) models that connect visual observations and language to actions. 3D-Mix [194] and GeoAware-VLA [1] use plug-and-play or implicit feature fusion, while GLaD [59] distills VGGT [167] features into the visual tokens of a VLA model. AugVLA-3D [132] augments action features with depth cues derived from VGGT [167]. ABot-M0 [191] admits VGGT [167] as one optional plug-and-play geometry module in a dual-stream policy, rather than making it the policy's core. VGGDrive [169] injects cross-view features into a driving VLM for risk perception, motion prediction, and trajectory planning, rather than direct low-level control. Controlled analysis [190] compares several VGGT [167]-VLA fusion strategies under matched settings. These models mainly use multi-view geometry latent features and retain structured geometric outputs when contact, collision, or feasibility requires geometric constraints.

Together, these directions use geometric state at 3 levels of embodied decision making: learned action prediction, explicit planning, and multimodal action reasoning. Multi-view geometry latent

Table 3. Evaluation-task coverage within the dataset collection summarized in Table 4. Each cell reports the number and percentage of datasets that provide observations or annotations supporting the corresponding evaluation target, with background color encoding the percentage.

Dataset group	Dataset count	State-strengthening tasks					Downstream-reuse tasks				
		Cam. pose	Depth	3D recon.	Long stream	Dyn. 4D	NVS	SLAM	Sem. under.	World models	Emb. action/plan.
3D reconstruction	46	36 (78%)	35 (76%)	24 (52%)	18 (39%)	15 (33%)	8 (17%)	20 (43%)	15 (33%)	5 (11%)	0 (0%)
NVS	10	4 (40%)	1 (10%)	1 (10%)	0 (0%)	2 (20%)	10 (100%)	0 (0%)	0 (0%)	2 (20%)	0 (0%)
Other downstream	15	1 (7%)	2 (13%)	1 (7%)	0 (0%)	2 (13%)	0 (0%)	0 (0%)	5 (33%)	8 (53%)	10 (67%)

Coverage 0% 100%

features support transferable policy representations, while structured geometric outputs provide the spatial precision needed to evaluate feasible motion and interaction.

4.6 Summary of Downstream State Reuse

Across the 5 downstream applications, the required access to geometric state depends on the capability being built. NVS and SLAM rely strongly on structured geometric outputs, semantic applications attach meaning to spatial anchors, world models require representations that support transitions beyond observed evidence, and embodied methods combine geometric state with action objectives. Although this is not a strict progression, applications closer to prediction and closed-loop action tend to combine the geometric precision of $\mathcal{R}_{\text{geo}}$ with the transferable information in $\mathcal{Z}_{\text{geo}}$. The common requirement is therefore not one fixed way of using the state, but preservation of the information needed by each application under its supervision and temporal setting; Section 5 next examines the evidence used to evaluate these capabilities.

5 Datasets and Benchmarks

This section reviews the datasets and evaluation targets used for geometric-state strengthening and downstream reuse. Section 5.1 groups datasets by their primary use in 3D reconstruction, NVS, or other downstream tasks. Sections 5.2 and 5.3 then examine how state-strengthening methods and downstream applications are evaluated, and Section 5.4 summarizes the limitations of current datasets and evaluation targets.

5.1 Dataset Taxonomy

We organize evaluation resources into 3 groups according to their primary use: datasets for 3D reconstruction, datasets for novel-view synthesis (NVS), and datasets for other downstream tasks. The assignment is not exclusive; a dataset remains in its principal group even when its observations or annotations support the evaluation of additional tasks. The collection includes datasets used by the reviewed methods and established resources for the corresponding task families. Table 3 first summarizes task-support coverage within this collection. Table 4 then provides dataset-level details, including scene type, source, reported dataset scale, and supported strengthening or reuse tasks.

Datasets for 3D reconstruction. The evolution of reconstruction datasets can be traced through 3 representative stages. DTU [73] established controlled multi-view evaluation on object-scale scans; ETH3D [141] later extended accurate geometry to high-resolution indoor and outdoor

scenes; and Waymo [152] now represents large-scale, multimodal outdoor data with broad task annotations. This evolution proceeds along 3 directions. First, spatial coverage expands from isolated objects to rooms, landmarks, and city-scale environments, as illustrated by ScanNet v2 [30], ScanNet200 [137], MegaDepth [94], and Waymo [152]. Second, temporal coverage moves from bounded image sets to trajectories and long sequences, with KITTI [52], TartanAir [177], and Waymo [152] supporting trajectory estimation and long-sequence reconstruction evaluation. Third, sensing conditions become more diverse: Replica [148] and Kubric [55] provide exact synthetic geometry and controllable motion, while real collections increasingly include changing illumination, non-pinhole cameras, events, thermal sensing, and LiDAR. Within the dataset collection summarized in Table 3, this group provides broader support for camera pose, depth, reconstruction, long-stream, dynamic, and SLAM evaluation.

Datasets for NVS. NVS benchmarks likewise progress from established video collections to larger and more diverse scene collections. RealEstate10K [221] provided an early large collection of posed real-estate videos; Mip-NeRF 360 [6] later established evaluation for unbounded scenes; and DL3DV-10K [97] now supplies large-scale real-world videos spanning indoor and outdoor environments. Their development follows 3 directions. First, scene coverage broadens across object-centric, outdoor, and unbounded settings: CO3Dv2 [133] covers 51 object categories through multi-view object-centric sequences, ACID [98] focuses on outdoor scenes, and Mip-NeRF 360 [6] evaluates unbounded environments. Second, temporal complexity extends static-scene rendering to dynamic content through DyCheck [47] and HyperNeRF [126]. Third, data scale increases from a small number of carefully captured scenes, such as Deep Blending [64] and LLFF [114], to large real-video collections such as RealEstate10K [221] and DL3DV-10K [97]. Table 3 shows that all datasets in this group support NVS evaluation, while fewer support depth, reconstructed geometry, temporal consistency, or downstream tasks.

Datasets for other downstream tasks. This group is organized by the applications being evaluated. Semantic datasets range from urban scene understanding in Cityscapes [29] to referring expressions in MVRefer [179], interaction-aware 4D understanding in HOI4D [102], and aligned egocentric and exocentric observations in Ego-Exo4D [54]. World-model datasets support interaction-conditioned or controllable future generation through HoI HQ [214] and Scene-Decoupled-Video-dataset [70]. Embodied datasets progress from controlled manipulation suites such as MetaWorld [196], CALVIN [113], ManiSkill 3 [159], LIBERO [99], and RoboTwin 2.0 [20] to real-world interaction and driving data in DROID [81], Ego-Exo4D [54], and NAVSIM [34]. Cityscapes [29] provides 5,000 urban video snippets, while NAVSIM [34] expands task-oriented evaluation to 115,000 filtered driving scenarios. Table 3 consequently shows moderate coverage for semantic understanding, broader coverage for world models, and the strongest coverage for embodied action, but little support for NVS or SLAM evaluation targets; these datasets measure application outcomes well but provide less evidence about whether gains arise specifically from improved geometric state.

5.2 Evaluation Targets for Geometric State Strengthening

Evaluation targets for diverse-input, efficient, robust, streaming and long-sequence, and dynamic reconstruction often assess the same geometric outputs under different conditions. Diverse-input methods vary the available measurements, efficient methods vary per-forward-pass precision and computation, and robust methods vary sensing conditions or input corruption. Streaming and long-sequence methods evaluate state transfer and alignment across windows, while dynamic methods evaluate time-varying scenes. Geometric accuracy must therefore be reported together with the condition being tested. Table 5 summarizes common geometric measures and the evaluation focus, representative datasets, key metrics, and comparison conditions for each strengthening direction.

Table 4. Datasets included in the evaluation taxonomy, grouped by primary use and supported tasks. A ✓ indicates that the dataset provides observations or annotations supporting the corresponding evaluation target; an ✗ indicates that such support was not identified. #Scenes reports the dataset scale used in this table; alternative reported scales are shown in parentheses when a scene count is unavailable. Type symbols denote  objects,  indoor scenes,  outdoor scenes, and  indoor and outdoor scenes. Source identifies real, synthetic, or mixed data.

Dataset	#Scenes	Type	Source	State-strengthening tasks					Downstream-reuse tasks				
				Cam. pose	Depth	3D recon.	Long stream	Dyn. 4D	NVS	SLAM	Sem. Under.	World models	Emb. action/plan.
Datasets for 3D Reconstruction													
DTU [73]	124		Real	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗
TUM-dynamics [149]	1		Real	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗
ADT [123]	2		Real	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗
EuRoC MAV [9]	3		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
TUM RGB-D [149]	4		Real	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
7-Scenes [144]	7		Real	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
NRGBD [5]	10		Synthetic	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
Replica360-2K [136]	13		Synthetic	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
Replica [148]	18		Synthetic	✓	✓	✓	✗	✗	✓	✓	✓	✗	✗
Bonn [119]	24		Real	✓	✓	✓	✓	✓	✗	✓	✗	✗	✗
Matterport3D [17]	90		Real	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗
Dark3R [58]	104		Real	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
Stanford2D3D [3]	271		Real	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗
NYU-v2 [146]	464		Real	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗
ScanNet++ v2 [192]	1,006		Real	✓	✓	✓	✗	✗	✓	✓	✓	✗	✗
ScanNet v2 [30]	1,513		Real	✓	✓	✓	✗	✗	✗	✓	✓	✗	✗
ScanNet200 [137]	1,513		Real	✓	✓	✓	✗	✗	✗	✓	✓	✗	✗
ARKitScenes [7]	1,661		Real	✓	✓	✓	✗	✗	✗	✗	✓	✗	✗
EventScape [51]	4		Synthetic	✗	✓	✗	✗	✓	✗	✗	✓	✗	✗
Nordland [155]	4		Real	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
Cambridge Landmarks [79]	5		Real	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗
Virtual KITTI 2 [11]	5		Synthetic	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗
Oxford Spires [160]	6		Real	✓	✗	✓	✓	✗	✓	✓	✗	✗	✗
DENSE [65]	8		Synthetic	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
KITTI-360 [95]	11		Real	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
PointOdyssey [216]	20		Synthetic	✓	✓	✗	✓	✓	✗	✗	✗	✗	✗
KITTI Odometry [52]	22		Real	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
TartanAir [177]	30		Synthetic	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
MegaDepth [94]	196		Real	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
nuScenes [13]	1,000		Real	✓	✓	✗	✗	✓	✓	✗	✓	✓	✗
Waymo [152]	1,150		Real	✓	✓	✗	✓	✓	✓	✗	✓	✓	✗
Oxford RobotCar [108]	– (~20M frames) ^a		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
LaMAR [138]	3		Real	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗
STheReO [199]	3		Real	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
FIORD [57]	10		Real	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
MVSEC [223]	11		Real	✓	✓	✗	✓	✗	✗	✓	✗	✗	✗
BEDLAM2.0 [161]	15		Synthetic	✓	✓	✗	✗	✓	✗	✗	✗	✗	✗
Tanks and Temples [83]	21		Real	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗
ETH3D [141]	25		Real	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
NCLT [16]	27		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
TUM-VI [142]	28		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
Sintel [10]	35		Synthetic	✓	✓	✗	✗	✓	✗	✗	✗	✗	✗

^aFor Oxford RobotCar [108], OmniWorld [222], and SPair-71k [116], we did not identify a reported scene count in the source used here; frame or pair counts are shown to indicate dataset scale.

Within bounded reconstruction, diverse-input targets vary intrinsics, poses, depth, partial geometry, sensing modality, or camera model and test whether these inputs improve camera, depth,

TABLE 4 (CONTINUED)

Dataset	#Scenes	Type	Source	State-strengthening tasks					Downstream-reuse tasks				
				Cam. pose	Depth	3D recon.	Long stream	Dyn. 4D	NVS	SLAM	Sem. Under.	World models	Emb. action/plan.
Datasets for 3D Reconstruction (continued)													
ViViD++ [86]	36	 	Real	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
DAVIS [128]	50	 	Real	✗	✗	✗	✗	✓	✓	✗	✓	✗	✗
Kubric [55]	10,000	 	Synthetic	✓	✓	✓	✗	✓	✗	✗	✗	✗	✗
OmniWorld [222]	— (18.515M frames)	 	Real + Synthetic	✓	✓	✓	✗	✓	✗	✗	✗	✓	✗
Datasets for NVS													
CO3Dv2 [133]	51		Real	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗
NVS-HO [2]	67		Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
Deep Blending [64]	19		Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
ACID [98]	891		Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
DyCheck [47]	7	 	Real	✓	✓	✗	✗	✓	✓	✗	✗	✗	✗
LLFF [114]	8	 	Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
Mip-NeRF 360 [6]	9	 	Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
HyperNeRF [126]	10	 	Real	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗
RealEstate10K [221]	~10,000	 	Real	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗
DL3DV-10K [97]	10,510	 	Real	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗
Datasets for other downstream tasks													
SPair-71k [116]	— (70,958 pairs)		Real	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
SIMPLER [93]	8		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
CALVIN [113]	34		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
MetaWorld [196]	50		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
RoboTwin 2.0 [20]	50		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
HOI4D [102]	57		Real	✗	✓	✓	✗	✓	✗	✗	✓	✗	✓
ManiSkill 3 [159]	58		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
DROID [81]	86		Real	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
LIBERO [99]	130		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
MVRefer [179]	1,513		Real	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
Cityscapes [29]	5,000		Real	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗
NAVSIM [34]	115,000		Real	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Ego-Exo4D [54]	131	 	Real	✓	✗	✗	✗	✓	✗	✗	✓	✓	✓
HolIHQ [214]	3,000	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗
Scene-Decoupled-Video-dataset [70]	3,400	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗

and point predictions. Efficiency targets retain the same geometric outputs while changing view count, resolution, numerical precision, token budget, architecture, or hardware; runtime, memory, computation, and geometric accuracy are then reported for a single forward-pass setting. Robustness targets instead vary the type and severity of visual degradation, environmental interference, occlusion, or transient distractors. Because the output tasks overlap, these conditions must be matched before accuracy, latency, memory, efficiency, or robustness gains are compared.

Temporal targets cover streaming, long-sequence, and dynamic reconstruction. Streaming and long-sequence evaluation combines trajectory drift and retained geometry with sequence and window length, cache or state size, chunk overlap, alignment, loop closure, revisit conditions, and whether observations are processed offline or causally. Dynamic targets further separate camera motion, static structure, moving regions, and dense 3D motion through trajectory, mask, flow, and reconstruction metrics. Reports must pair each temporal target with the geometry retained over the evaluated sequence.

Table 5. Evaluation for geometric state strengthening. Each group lists its evaluation focus, representative datasets, key metrics, and comparison conditions. Common geometric measures are summarized above the groups.

Evaluation focus	Representative datasets	Key metrics	Key comparison conditions
Common geometry metrics, selected by task:			
Pose: relative rotation / translation accuracy (RRA / RTA), pose area under the curve (AUC).			
Depth: absolute relative error (AbsRel), root mean squared error (RMSE), threshold accuracy at 1.25 (δ_1).			
Surface: Chamfer distance (CD), accuracy, completeness.			
Diverse-input 3D reconstruction			
Geometry with diverse inputs	Pose: CO3Dv2 [133] Depth: NYU-v2 [146] Surface: DTU [73] Multimodal: ViViD++ [86] Event: MVSEC [223] Panoramic: Replica360-2K [136]	Pose / depth / surface metrics for the supplied inputs	Input modality / sparsity / noise; camera model / calibration; view count / metric scale; backbone adaptation
Efficient 3D reconstruction			
Geometry quality and computation	Pose: CO3Dv2 [133] Geometry: ETH3D [141]; ScanNet [30]	Geometry metrics; FPS / latency; peak GPU memory; FLOPs / bits per weight	Views / resolution / batch size; precision / token budget; architecture / hardware; timing scope
Robust 3D reconstruction			
Geometry under adverse conditions or outlier views	Adverse conditions: Dark3R [58]; FIORD [57] Outlier views: ETH3D [141]	Depth: AbsRel / $\text{RMSE}_{\log}$ / δ_1 ; pose: RRA / RTA / AUC; rotation / translation-direction error	Environment / degradation severity; distractor ratio / view overlap; paired clean and degraded inputs; adaptation
Streaming and long-sequence 3D reconstruction			
Trajectory drift, retained geometry, and resource use	Trajectory: TUM RGB-D [149]; KITTI Odometry [52] Long sequences: TartanAir [177]; Waymo [152]	Geometry metrics; ATE / RPE; translation / rotation drift; FPS / peak GPU memory	Offline / causal processing; sequence / window length; chunk overlap / stride; memory budget / warm-up; alignment / loop closure / revisits
Dynamic 3D reconstruction			
Dynamic geometry, motion, and segmentation	Camera / depth: Bonn [119] Motion: PointOdyssey [216]; Kubric [55] Masks / views: DAVIS [128]; DyCheck [47]	Geometry metrics / ATE; 3D endpoint error; mask IoU / boundary F-score	Static-dynamic split; frame span / camera motion; mask availability; scale / temporal alignment

ATE: absolute trajectory error; RPE: relative pose error; $\text{RMSE}_{\log}$: RMSE of log-depth; FPS: frames per second; FLOPs: floating-point operations; IoU: intersection over union.

5.3 Evaluation Targets for Downstream State Reuse

Current downstream targets measure the resulting application capability rather than the contribution of the geometric-state information used by the application. Because the 5 applications produce different outputs, we examine their targets directly in the order of NVS, SLAM, semantic 3D scene understanding, world models, and embodied applications. Table 6 links evaluation tasks to representative datasets, key metrics, and comparison conditions within each application group.

Table 6. Evaluation for downstream geometric-state reuse. Rows link evaluation tasks to representative datasets, key metrics, and comparison conditions within each application group.

Evaluation task	Representative datasets	Key metrics	Key comparison conditions
NVS			
Static-scene NVS	RealEstate10K [221]; DL3DV-10K [97] CO3Dv2 [133]; Mip-NeRF 360 [6]	PSNR / SSIM / LPIPS	Input / target views; cameras / resolution; scene-specific optimization
Dynamic-scene NVS	DAVIS [128]; DyCheck [47]; HyperNeRF [126]	PSNR / SSIM / LPIPS	View / temporal split; masks / dynamic content; scene-specific optimization
SLAM			
Tracking and mapping	TUM RGB-D [149]; EuRoC [9]	Trajectory: ATE / RPE; map: accuracy / completeness	Trajectory / alignment; loop closure / map retention
Visual odometry	KITTI [52]; TUM-VI [142]	ATE / RPE; translation / rotation drift	Trajectory length; scale / pose alignment
Relocalization	Cambridge Landmarks [79]; LaMAR [138]	Translation / rotation error; localization recall	Query / reference split; map / pose thresholds
Place recognition	NCLT [16]; Oxford RobotCar [108]	Recall@K	Query / retrieval database; matching threshold / K
Semantic 3D scene understanding			
3D semantic prediction and mapping	ScanNet200 [137]; ARKitScenes [7]	Prediction: mIoU / AP	Class vocabulary; 2D–3D label mapping
	SPair-71k [116]	Correspondence: PCK	View split; keypoint threshold
Language-based 3D scene reasoning	MVRefer [179]	Grounding accuracy at IoU 0.25	Language queries; view split; grounding threshold
Geometry-aware world models			
Video-depth prediction	Cityscapes [29]; TartanAir [177]	AbsRel / RMSE	Prediction horizon; camera conditioning
Video generation	HolHQ [214]; Scene-Decoupled-Video-dataset [70]	Visual: PSNR / SSIM / LPIPS / FVD; geometry: reprojection error / PCK	Camera / interaction conditions; sampling budget; teacher forcing
Action prediction / rollout selection	DROID [81]; LIBERO [99]	Task success rate	Action conditioning; horizon / rollout budget; selection policy
Embodied action and planning			
Manipulation	ManiSkill 3 [159]; RoboTwin 2.0 [20]; DROID [81]	Task success rate	Task suite / demonstrations; cameras / control horizon
Long-horizon policies / generalization	LIBERO [99]; CALVIN [113]; SIMPLER [93]	Task success rate	Task sequence / test conditions; policy initialization; encoder adaptation
Driving	NAVSIM [34]	PDM score	Scenario split; camera setup / control horizon

PSNR: peak signal-to-noise ratio; SSIM: structural similarity index measure; LPIPS: learned perceptual image patch similarity; FVD: Fréchet video distance; mIoU: mean intersection over union; AP: average precision; PCK: percentage of correct keypoints. Geometric metric abbreviations follow Table 5.

NVS targets use held-out views to measure appearance and, where geometry is available, surface fidelity. Input and target views, camera information, resolution, scene-specific optimization, masks, and dynamic content must be aligned before rendering or geometry scores are compared. High rendering quality alone does not establish accurate surfaces.

SLAM targets combine trajectory and map evaluation with retrieval and metric-pose evaluation for relocalization. Trajectory segments, alignment rules, loop-closure settings, map retention, retrieval databases, and decision thresholds must be matched; low local trajectory drift can otherwise coexist with global map loss.

Semantic-understanding targets evaluate objects, regions, occupancy, correspondence, cross-epoch change, language grounding, or spatial reasoning. Because geometry is assessed through task-specific annotations, comparisons must align class vocabularies, 2D–3D label mappings, view and temporal splits, registrations, query sets, and matching thresholds.

World-model targets extend observed geometry to future observations or actions. They assess geometry forecasting, future-view or video generation, and action-conditioned rollouts using geometric consistency, visual fidelity, and, where applicable, task success. Prediction horizon, conditioning, sample budget, and rollout-selection rule must be specified because a visually plausible single-step future does not establish persistent geometry or useful long-horizon state transitions.

Embodied targets report task or manipulation success rates, path efficiency, collision rates, driving comfort, or trajectory quality. Geometry is evaluated indirectly through action outcomes, so the task suite, demonstrations, camera setup, control horizon, policy initialization, and encoder adaptation must be fixed before gains are attributed to the geometric-state information supplied to the application.

Across these applications, downstream scores—such as segmentation accuracy, video quality, manipulation success rate, and trajectory quality—characterize the resulting capability. The next subsection summarizes the observed limitations in dataset coverage and target comparability.

5.4 Limitations of Current Datasets and Evaluation Targets

The reviewed tasks do not have equal evaluation coverage. Static RGB geometry and NVS have comparatively mature datasets and targets, whereas long streams, dynamic scenes, specialized sensors, semantic persistence, embodied interaction, and world-model prediction rely on smaller or task-specific collections. Few datasets contain the geometric, temporal, semantic, and action annotations needed to evaluate several capabilities on the same scenes.

Evaluation targets also vary within a dataset. Input count, image resolution, temporal window, train–test split, scale alignment, model adaptation, and hardware are not standardized across papers. Moreover, current targets rarely test what geometric information $\mathcal{Z}_{\text{geo}}$ retains, and cameras, depth maps, point maps, and tracks in $\mathcal{R}_{\text{geo}}$ are often scored separately without checking whether they describe consistent geometry. Dataset coverage is therefore uneven, and results obtained under different target settings are not directly comparable.

6 Open Challenges

The preceding sections show that progress is uneven. Feed-forward reconstruction is well established for bounded, predominantly static image sets, and later methods address diverse inputs, resource limits, difficult sensing, long streams, and scene motion. Reuse is also well developed for rendering and geometric mapping, but remains less established for semantic, geometry-aware world-model, and embodied applications. The reviewed evidence exposes 3 field-level problems concerning the reliability of geometric state, its adaptation to different applications, and dataset and benchmark design for identifying the contribution of geometric state:

- How can geometric state remain reliable and persistent as measurements, sequence length, and scene motion change?
- How can geometric representations adapt to the different information required by downstream applications?
- How should datasets and benchmarks be designed to determine whether application gains result from changes in or reuse of geometric state?

The 3 subsections below discuss these problems in turn, summarizing current evidence, the remaining gap, and promising research directions.

6.1 Reliable and Persistent Geometric State

Recent work has strengthened feed-forward reconstruction in several aspects. Diverse-input models accept camera parameters, depth, partial geometry, LiDAR, or non-standard visual inputs; robust models address degraded and ambiguous visual evidence; efficient models reduce the cost of a forward pass; streaming and long-sequence models transfer and align state across windows; and dynamic models represent scene motion. These advances extend $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$ beyond clean, fixed RGB image sets, but they do not yet provide a single state that remains reliable as sensing conditions, scene extent, and time change together.

The first problem is that a supplied measurement is not guaranteed to remain in the final geometry. In many diverse-input models, known depth, metric scale, calibration, or LiDAR samples are encoded into $\mathcal{Z}_{\text{geo}}$ and decoded through learned heads. The resulting structured geometric outputs can improve without showing which measured values were preserved, which regions were completed by the model, or which predictions are unreliable. MapAnything [78] and LiDAR-VGGT [171] make geometric constraints more explicit, but they do not establish a common test for measurement preservation and uncertainty.

The second problem is that no reviewed system simultaneously handles bounded computation and memory, a stable global coordinate frame, reliable recognition of revisited places, and persistent identities for moving objects as a sequence grows. StreamVGGT [224] and FrameVGGT [186] explore cached or bounded short-term context, Anchor3R [158] introduces transient anchors for long-horizon alignment, and PAGE-4D [220] and DynamicVGGT [63] model scene motion. These components remain separate, however, and have not been unified with the diverse sensing settings reviewed in Section 3.1. A short-term cache may forget global anchors, a compact state may discard observations needed later, and a static map may absorb moving objects into inconsistent geometry. The open question is how a single system can retain measured geometry, maintain global consistency, and separate persistent structure from change over long streams.

A useful direction is to retain measurement-linked geometry with explicit uncertainty and provenance, together with short-term information for current reconstruction, long-term anchors for scale and revisits, and object records for identity and motion. These components should remain geometrically aligned while being updated at different rates.

6.2 Adaptive Geometric Representations for Downstream Applications

The 5 application groups need different parts of geometric state. NVS uses surfaces, viewpoints, and appearance; SLAM uses a stable coordinate frame, trajectories, and long-term anchors; semantic applications attach objects and language to 3D locations; world models use state that changes with time and action; and embodied applications need free space, contact geometry, and reliability for action. Existing systems show that structured geometric outputs, multi-view geometry latent features, or their combination can support each individual requirement.

The problem is that these choices are made separately for each application. Some methods use one decoded output, whereas others use multi-view geometry latent features or distilled features, and their comparisons use different architectures and targets. Passing every feature and output is costly, while discarding information can remove what another application needs. There is still no common geometric state that can expose the information needed by different applications without rebuilding a separate representation for each application.

A useful direction is an adaptable state with stable geometric anchors and task-specific access: surfaces and cameras for rendering, camera trajectories and persistent landmarks for mapping, object-linked geometry for semantics, local constraints for action, and time-varying entities for prediction. Matched comparisons of $\mathcal{R}_{\text{geo}}$, $\mathcal{Z}_{\text{geo}}$, and their combination should test which parts are needed for each capability under the same data, supervision, and model capacity. The goal is a common state that applications can select and adapt to their needs.

6.3 Dataset Design for Benchmarking Geometric State

The central evaluation problem is whether an application gain actually comes from geometric-state reuse. A higher rendering, mapping, semantic, prediction, or action score does not answer this question when a method also changes the application architecture, adds supervision, increases model capacity, or introduces task-specific adaptation. Current results therefore establish that the complete application improved, but often cannot isolate the contribution of $\mathcal{Z}_{\text{geo}}$ or $\mathcal{R}_{\text{geo}}$.

A direct test requires a controlled comparison within the same application. The application model, training data, supervision, and training budget remain unchanged, while its input is replaced in turn by $\mathcal{R}_{\text{geo}}$, $\mathcal{Z}_{\text{geo}}$, their combination, and a baseline without VGGT geometric state. If an application improves only when geometric-state information is provided, the performance difference can be attributed to that information more directly. This comparison also reveals whether the application needs structured geometric outputs, multi-view geometry latent features, or their combination.

A benchmark for this purpose must make the controlled comparison reproducible. It needs a fixed data split, a fixed application model and training recipe, and evaluation targets for both geometry and the application outcome. Each geometric-state input is then tested under the same evaluation setup, and the benchmark reports both the change in geometric quality and the corresponding change in application performance. This design would show whether improving or using the geometric state leads to a measurable application gain, rather than merely showing that two different complete systems obtain different scores.

7 Conclusion

This survey organizes the VGGT [167] lineage around one geometric state with two components: the multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$ and the structured geometric outputs $\mathcal{R}_{\text{geo}}$. State-strengthening work improves how this state is formed from diverse inputs, computed efficiently, made robust, retained over streams, and extended to dynamic 3D reconstruction. State-reuse work uses structured geometric outputs, multi-view geometry latent features, or their combination for NVS, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning.

The dataset and benchmark analysis reveals uneven evaluation coverage: static reconstruction and rendering are well supported, whereas long-stream, dynamic, and task-oriented settings rely more heavily on specialized resources. Open challenges include maintaining reliable and persistent geometric state, adapting representations to different applications, and designing datasets and benchmarks that isolate whether application gains arise from changes in or reuse of geometric state. Addressing these challenges can extend VGGT [167]-based geometric state from isolated reconstruction outputs to broader applications.

References

- [1] Ali Abouzeid, Malak Mansour, Qinbo Sun, Zezhou Sun, and Dezhen Song. 2025. GeoAware-VLA: Implicit Geometry Aware Vision-Language-Action Model. *arXiv preprint arXiv:2509.14117* (2025). <https://arxiv.org/abs/2509.14117>
- [2] Musawar Ali, Manuel Carranza-Garcia, Nicola Fioraio, Samuele Salti, and Luigi Di Stefano. 2026. NVS-HO: A Benchmark for Novel View Synthesis of Handheld Objects. *arXiv preprint arXiv:2602.05822* (2026). <https://arxiv.org/abs/2602.05822>
- [3] Iro Armeni, Sasha Sax, Amir R. Zamir, and Silvio Savarese. 2017. Joint 2D–3D–Semantic Data for Indoor Scene Understanding. *arXiv preprint arXiv:1702.01105* (2017). <https://arxiv.org/abs/1702.01105>
- [4] Alaa Asfour, Christopher Indris, Lei Han Chen, Tejas Vyas, and Guanghui Wang. 2026. Distilling 3D Spatial Reasoning into a Lightweight Vision-Language Model with CoT. *arXiv preprint arXiv:2605.09719* (2026). <https://arxiv.org/abs/2605.09719>
- [5] Dejan Azinović, Ricardo Martin-Brualla, Dan B. Goldman, Matthias Nießner, and Justus Thies. 2022. Neural RGB-D Surface Reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr52688.2022.00619
- [6] Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2022. Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance Fields. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr52688.2022.00539
- [7] Gilad Baruch, Zhuoyuan Chen, Afshin Dehghan, Tal Dimry, Yuri Feigin, Peter Fu, Thomas Gebauer, Brandon Joffe, Daniel Kurz, Arik Schwartz, and Elad Shulman. 2021. ARKitScenes: A Diverse Real-World Dataset for 3D Indoor Scene Understanding Using Mobile RGB-D Data. In *Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS Datasets and Benchmarks)*. <https://arxiv.org/abs/2111.08897>
- [8] Jelena Bratulić, Sudhanshu Mittal, Thomas Brox, and Christian Rupprecht. 2025. On Geometric Understanding and Learned Priors in Feed-Forward 3D Reconstruction Models. *arXiv preprint arXiv:2512.11508* (2025). <https://arxiv.org/abs/2512.11508>
- [9] Michael Burri, Janosch Nikolic, Pascal Gohl, Thomas Schneider, Joern Rehder, Sammy Omari, Markus W. Achtelik, and Roland Siegwart. 2016. The EuRoC Micro Aerial Vehicle Datasets. *The International Journal of Robotics Research (IJRR)* (2016). doi:10.1177/0278364915620033
- [10] Daniel J. Butler, Jonas Wulff, Garrett B. Stanley, and Michael J. Black. 2012. A Naturalistic Open Source Movie for Optical Flow Evaluation. In *European Conference on Computer Vision (ECCV)*. doi:10.1007/978-3-642-33783-3_44
- [11] Johann Cabon, Naila Murray, and Martin Humenberger. 2020. Virtual KITTI 2. *arXiv preprint arXiv:2001.10773* (2020). <https://arxiv.org/abs/2001.10773>
- [12] Johann Cabon, Lucas Stofl, Leonid Antsfeld, Gabriela Csurka, Boris Chidlovskii, Jerome Revaud, and Vincent Leroy. 2025. MUST3R: Multi-view Network for Stereo 3D Reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1050–1060. doi:10.1109/CVPR52734.2025.00106
- [13] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. 2020. nuScenes: A Multimodal Dataset for Autonomous Driving. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr42600.2020.01164
- [14] Yifei Cao, Yu Liu, Guolong Wang, Zhu Liu, Kai Wang, Xianjie Zhang, Jizhe Yu, and Xun Tu. 2026. EAGLE: Episodic Appearance- and Geometry-Aware Memory for Unified 2D–3D Visual Query Localization in Egocentric Vision. In *AAAI Conference on Artificial Intelligence (AAAI)*. <https://arxiv.org/abs/2511.08007>
- [15] Yang Cao, Feize Wu, Dave Zhenyu Chen, Yingji Zhong, Lanqing Hong, and Dan Xu. 2026. VGGT-Det: Mining VGGT Internal Priors for Sensor-Geometry-Free Multi-View Indoor 3D Object Detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 4708–4717. https://openaccess.thecvf.com/content/CVPR2026/html/Cao_VGGT-Det_Mining_VGGT_Internal_Priors_for_Sensor-Geometry-Free_Multi-View_Indoor_3D_CVPR_2026_paper.html
- [16] Nicholas Carlevaris-Bianco, Arash K. Ushani, and Ryan M. Eustice. 2016. University of Michigan North Campus Long-Term Vision and LiDAR Dataset. *The International Journal of Robotics Research (IJRR)* (2016). doi:10.1177/0278364915614638
- [17] Angel X. Chang, Angela Dai, Thomas Funkhouser, Matthias Nießner, Manolis Savva, Shuran Song, Andy Zeng, and Yinda Zhang. 2017. Matterport3D: Learning from RGB-D Data in Indoor Environments. In *International Conference on 3D Vision (3DV)*. doi:10.1109/3dv.2017.00081
- [18] Guikun Chen and Wenguan Wang. 2026. A Survey on 3D Gaussian Splatting. *ACM Computing Surveys (CSUR)* 58, 12 (2026), 1–39. doi:10.1145/3807511
- [19] Ruiyang Chen, Feiran Li, Chu Zhou, Zonglin Li, Zhanyu Ma, and Heng Guo. 2026. OF³GS: On-the-Fly Feed-Forward 3D Gaussian Splatting from Unposed Images. *arXiv preprint arXiv:2606.03254* (2026). <https://arxiv.org/abs/2606.03254>
- [20] Tianxing Chen, Zhanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, Weiliang Deng, Yubin Guo, Tian Nian, Xuanbing Xie, Qiangyu Chen, Kailun Su, Tianling Xu, Guodong Liu,

- Mengkang Hu, Huan ang Gao, Kaixuan Wang, Zhixuan Liang, Yusen Qin, Xiaokang Yang, Ping Luo, and Yao Mu. 2026. RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation. In *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2506.18088>
- [21] Tianrun Chen, Yuanqi Hu, Yidong Han, Hanjie Xu, Deyi Ji, Qi Zhu, Chunan Yu, Xin Zhang, Cheng Chen, Chaotao Ding, Ying Zang, Xuanfu Li, Jin Ma, and Lanyun Zhu. 2026. HD-VGGT: High-Resolution Visual Geometry Transformer. *arXiv preprint arXiv:2603.27222* (2026). <https://arxiv.org/abs/2603.27222>
- [22] Xun Chen, Tianchen Deng, Rui Wang, Fangjinhua Wang, Junyi Ma, Hongming Shen, Hesheng Wang, and Danwei Wang. 2026. VGGT-Occ: Geometry-Grounded and Density-Aware Gated Fusion for 3D Occupancy Prediction. *arXiv preprint arXiv:2605.16911* (2026). <https://arxiv.org/abs/2605.16911>
- [23] Yutian Chen, Yuheng Qiu, Ruogo Li, Jay Patrikar, and Sebastian Scherer. 2026. Co-Me: Confidence-Guided Token Merging for Visual Geometric Transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14590–14599. https://openaccess.thecvf.com/content/CVPR2026/html/Chen_Co-Me_Confidence_Guided_Token_Merging_for_Visual_Geometric_Transformers_CVPR_2026_paper.html
- [24] Ziyang Chen, Yansong Qu, You Shen, Xuan Cheng, and Liujuan Cao. 2026. StereoVGGT: A Training-Free Visual Geometry Transformer for Stereo Vision. *arXiv preprint arXiv:2603.29368* (2026). <https://arxiv.org/abs/2603.29368>
- [25] Zhangquan Chen, Manyuan Zhang, Xinlei Yu, Xufang Luo, Mingze Sun, Zihao Pan, Xiang An, Yan Feng, Peng Pei, Xunliang Cai, and Ruqi Huang. 2026. Think with 3D: Geometric Imagination Grounded Spatial Reasoning from Limited Views. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2510.18632>
- [26] Chong Cheng, Xianda Chen, Tao Xie, Wei Yin, Weiqiang Ren, Qian Zhang, Xiaoyang Guo, and Hao Wang. 2026. LongStream: Long-Sequence Streaming Autoregressive Visual Geometry. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. https://openaccess.thecvf.com/content/CVPR2026/html/Cheng_LongStream_Long-Sequence_Streaming_Autoregressive_Visual_Geometry_CVPR_2026_paper.html
- [27] Jiho Choi, Seonho Lee, Seojeong Park, and Hyunjung Shim. 2026. Dense Reward for Multi-View 3D Reasoning with Global Maps and Local Views. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2606.23557>
- [28] Jean Cordonnier, Chenghao Xu, Olga Fink, and Malcolm Mielle. 2026. Unpaired RGB-Thermal Gaussian-Splatting Using Visual Geometric Transformers. In *MM-SpatialAI Workshop at the IEEE International Conference on Robotics and Automation (ICRA Workshop)*. <https://arxiv.org/abs/2606.05491>
- [29] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. 2016. The Cityscapes Dataset for Semantic Urban Scene Understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2016.350
- [30] Angela Dai, Angel X. Chang, Manolis Savva, Maciej Halber, Thomas Funkhouser, and Matthias Nießner. 2017. ScanNet: Richly-Annotated 3D Reconstructions of Indoor Scenes. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2017.261
- [31] Jun Dai, Renbiao Jin, Bo Xu, Yutian Chen, Linning Xu, Mulin Yu, Tianfan Xue, and Shi Guo. 2026. AsyncEvGS: Asynchronous Event-Assisted Gaussian Splatting for Handheld Motion-Blurred Scenes. *arXiv preprint arXiv:2605.07192* (2026). <https://arxiv.org/abs/2605.07192>
- [32] Weichen Dai, Wenhan Su, Da Kong, Yuhang Ming, and Wanzeng Kong. 2026. Keyframe-Based Feed-Forward Visual Odometry. *arXiv preprint arXiv:2601.16020* (2026). <https://arxiv.org/abs/2601.16020>
- [33] Nobel Dang and Bing Li. 2026. MultiLoc: Multi-view Guided Relative Pose Regression for Fast and Robust Visual Re-Localization. *arXiv preprint arXiv:2603.27170* (2026). <https://arxiv.org/abs/2603.27170>
- [34] Daniel Dauner, Marcel Hallgarten, Tianyu Li, Xinshuo Weng, Zhiyu Huang, Zetong Yang, Hongyang Li, Igor Gilitschenski, Boris Ivanovic, Marco Pavone, Andreas Geiger, and Kashyap Chitta. 2024. NAVSIM: Data-Driven Non-Reactive Autonomous Vehicle Simulation and Benchmarking. In *Advances in Neural Information Processing Systems (NeurIPS)*. doi:10.52202/079017-0902
- [35] Jianing Deng, Yuanzhe Li, Jialu Wang, Song Wang, Tianlong Chen, Huanrui Yang, and Jingtong Hu. 2026. SAF3R: Dynamic Sparse Attention for Feed-Forward 3D Reconstruction Transformers. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2607.03612>
- [36] Kai Deng, Zexin Ti, Jiawei Xu, Jian Yang, and Jin Xie. 2026. VGGT-Long: Chunk it, Loop it, Align it – Pushing VGGT’s Limits on Kilometer-scale Long RGB Sequences. In *IEEE International Conference on Robotics and Automation (ICRA)*. <https://arxiv.org/abs/2507.16443>
- [37] Tianchen Deng, Xun Chen, Ziming Li, Hongming Shen, Shuhao Zhai, Danwei Wang, Javier Civera, and Hesheng Wang. 2026. UniPR-3D: Towards Universal Visual Place Recognition with Visual Geometry Grounded Transformer. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2512.21078>
- [38] Tianchen Deng, Wenhua Wu, Kunzhen Wu, Guangming Wang, Siting Zhu, Shenghai Yuan, Xun Chen, Guole Shen, Zhe Liu, and Hesheng Wang. 2025. Reloc-VGGT: Visual Re-localization with Geometry Grounded Transformer. *arXiv preprint arXiv:2512.21883* (2025). <https://arxiv.org/abs/2512.21883>

- [39] Tianchen Deng, Zhenxiang Xiong, Nailin Wang, Fangjinhua Wang, Jiuming Liu, Jianfei Yang, and Hesheng Wang. 2026. Mamba-VGGT: Persistent Long-Sequence Video Geometry Grounded Transformer via External Sliding Window Mamba Memory. *arXiv preprint arXiv:2605.17478* (2026). <https://arxiv.org/abs/2605.17478>
- [40] Tianye Ding, Yiming Xie, Yiqing Liang, Moitreyia Chatterjee, Pedro Miraldo, and Huaizu Jiang. 2026. LASER: Layer-wise Scale Alignment for Training-Free Streaming 4D Reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2512.13680>
- [41] Hongyang Du, Junjie Ye, Xiaoyan Cong, Runhao Li, Jingcheng Ni, Aman Agarwal, Zeqi Zhou, Zekun Li, Randall Balestriero, and Yue Wang. 2026. VideoGPA: Distilling Geometry Priors for 3D-Consistent Video Generation. In *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2601.23286>
- [42] Ruxiao Duan, Erin Hong, Dongxu Zhao, Eric Turner, Alex Wong, and Yunwen Zhou. 2026. Fisheye3R: Adapting Unified 3D Feed-Forward Foundation Models to Fisheye Lenses. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2603.28896>
- [43] Sven Elflein, Ruilong Li, Sérgio Agostinho, Zan Gojcic, Laura Leal-Taixé, Qunjie Zhou, and Aljosa Osep. 2026. VGG-T³: Offline Feed-Forward 3D Reconstruction at Scale. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 36464–36474. https://openaccess.thecvf.com/content/CVPR2026/html/Elfleyn_VGG-T3_Offline_Feed-Forward_3D_Reconstruction_at_Scale_CVPR_2026_paper.html
- [44] Ningjing Fan and Yiqun Wang. 2026. SSR-GS: Separating Specular Reflection in Gaussian Splatting for Glossy Surface Reconstruction. *arXiv preprint arXiv:2603.05152* (2026). <https://arxiv.org/abs/2603.05152>
- [45] Denis Fatykhov, Timur Akhtyamov, Konstantin Pakulev, German Devchich, and Gonzalo Ferrer. 2026. MuViSeg: Multi-View Segment Correspondences from Dense Geometry Priors. *arXiv preprint arXiv:2607.17938* (2026). <https://arxiv.org/abs/2607.17938>
- [46] Weilun Feng, Haotong Qin, Mingqiang Wu, Chuanguang Yang, Yuqi Li, Xiangqi Li, Zhulin An, Libo Huang, Yulun Zhang, Michele Magno, and Yongjun Xu. 2026. Quantized Visual Geometry Grounded Transformer. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2509.21302>
- [47] Hang Gao, Ruilong Li, Shubham Tulsiani, Bryan Russell, and Angjoo Kanazawa. 2022. Monocular Dynamic View Synthesis: A Reality Check. In *Advances in Neural Information Processing Systems (NeurIPS)*. doi:10.52202/068431-2447
- [48] Yang Gao, Wuyang Li, Po-Chien Luan, and Alexandre Alahi. 2026. Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2605.28587>
- [49] Yulu Gao, Bohao Zhang, Zongheng Tang, Jitong Liao, Wenjun Wu, and Si Liu. 2026. VGGT-Segmentor: Geometry-Enhanced Cross-View Segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 21690–21700. https://openaccess.thecvf.com/content/CVPR2026/html/Gao_VGGT-Segmentor_Geometry-Enhanced_Cross-View_Segmentation_CVPR_2026_paper.html
- [50] Shijia Ge, Yinxin Zhang, Shuzhao Xie, Weixiang Zhang, Mingcai Zhou, and Zhi Wang. 2026. VGGT-DP: Generalizable Robot Control via Vision Foundation Models. In *AAAI Conference on Artificial Intelligence (AAAI)*. <https://arxiv.org/abs/2509.18778>
- [51] Daniel Gehrig, Michelle Rüegg, Mathias Gehrig, Javier Hidalgo-Carrió, and Davide Scaramuzza. 2021. Combining Events and Frames Using Recurrent Asynchronous Multimodal Networks for Monocular Depth Prediction. *IEEE Robotics and Automation Letters (RA-L)* (2021). doi:10.1109/lra.2021.3060707
- [52] Andreas Geiger, Philip Lenz, and Raquel Urtasun. 2012. Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2012.6248074
- [53] Anna Gelencsér-Horváth, Gergely Dinya, Dorka Boglárka Erős, Péter Halász, Islam Muhammad Muqsit, and Kristóf Karacs. 2026. SceneVGGT: VGGT-Based Online 3D Semantic SLAM for Indoor Scene Understanding and Navigation. In *IEEE International Conference on Image Processing (ICIP)*. <https://github.com/HBVC-AI/SceneVGGT>
- [54] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, Bikram Boote, Eugene Byrne, Zach Chavis, Joya Chen, Feng Cheng, Fugen Chu, Sean Crane, Avijit Dasgupta, Jing Dong, Maria Escobar, Crisithian Forigua, Abrham Gebreselasie, Sanjay Haresh, Jing Huang, Md Mohaiminul Islam, Suyog Jain, Rawal Khirodkar, Devansh Kukreja, Kevin J. Liang, Jia-Wei Liu, Sagnik Majumder, Yongsan Mao, Miguel Martin, Effrosyni Mavroudi, Tushar Nagarajan, Francesco Ragusa, Santhosh Kumar Ramakrishnan, Luigi Seminara, Arjun Somayazulu, Yale Song, Shan Su, Zihui Xue, Edward Zhang, Jinxu Zhang, Angela Castillo, Changan Chen, Xinzhu Fu, Ryosuke Furuta, Cristina González, Prince Gupta, Jiabo Hu, Yifei Huang, Yiming Huang, Weslie Khoo, Anush Kumar, Robert Kuo, Sach Lakhavani, Miao Liu, Mi Luo, Zhengyi Luo, Brighid Meredith, Austin Miller, Oluwatumininu Oguntola, Xiaqing Pan, Penny Peng, Shraman Pramanick, Merey Ramazanov, Fiona Ryan, Wei Shan, Kiran Somasundaram, Chenan Song, Audrey Southerland, Masatoshi Tateno, Huiyu Wang, Yuchen Wang, Takuma Yagi, Mingfei Yan, Xitong Yang, Zecheng Yu, Shengxin Cindy Zha, Chen Zhao, Ziwei Zhao, Zhifan Zhu, Jeff Zhuo, Pablo Arbeláez, Gedas Bertasius, David Crandall, Dima Damen, Jakob Engel,

- Giovanni Maria Farinella, Antonino Furnari, Bernard Ghanem, Judy Hoffman, C. V. Jawahar, Richard Newcombe, Hyun Soo Park, James M. Rehg, Yoichi Sato, Manolis Savva, Jianbo Shi, Mike Zheng Shou, and Michael Wray. 2025. Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives. *International Journal of Computer Vision (IJCV)* 133, 12 (2025), 8356–8435. doi:10.1007/s11263-025-02557-6
- [55] Klaus Greff, Francois Belletti, Lucas Beyer, Carl Doersch, Yilun Du, Daniel Duckworth, David J. Fleet, Dan Gnanaprasagam, Florian Golemo, Charles Herrmann, Thomas Kipf, Abhijit Kundu, Dmitry Lagun, Issam Laradji, Hsueh-Ti Liu, Henning Meyer, Yishu Miao, Derek Nowrouzezahrai, Cengiz Oztireli, Etienne Pot, Noha Radwan, Daniel Rebain, Sara Sabour, Mehdi S. M. Sajjadi, Matan Sela, Vincent Sitzmann, Austin Stone, Deqing Sun, Suhani Vora, Ziyu Wang, Tianhao Wu, Kwang Moo Yi, Fangcheng Zhong, and Andrea Tagliasacchi. 2022. Kubric: A Scalable Dataset Generator. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 3749–3761. https://openaccess.thecvf.com/content/CVPR2022/html/Greff_Kubric_A_Scalable_Dataset_Generator_CVPR_2022_paper.html
- [56] Yi Gu, Zhaorui Wang, Jiahang Cao, Jiaxu Wang, Mingle Zhao, Dongjun Ye, and Renjing Xu. 2026. Sparse View Distractor-Free Gaussian Splatting. *arXiv preprint arXiv:2603.01603* (2026). <https://arxiv.org/abs/2603.01603>
- [57] Ulas Gunes, Matias Turkulainen, Xuqian Ren, Arno Solin, Juho Kannala, and Esa Rahtu. 2025. FIORD: A Fisheye Indoor-Outdoor Dataset with LiDAR Ground Truth for 3D Scene Reconstruction and Benchmarking. In *Scandinavian Conference on Image Analysis (SCIA)*. 3–17.
- [58] Andrew Y. Guo, Anagh Malik, SaiKiran Tedla, Yutong Dai, Yiqian Qin, Zach Salehe, Benjamin Attal, Sotiris Nouisias, Kiriakos N. Kutulakos, and David B. Lindell. 2026. Dark3R: Learning Structure from Motion in the Dark. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 34160–34170. https://openaccess.thecvf.com/content/CVPR2026/html/Guo_Dark3R_Learning_Structure_from_Motion_in_the_Dark_CVPR_2026_paper.html
- [59] Minghao Guo, Meng Cao, Jiachen Tao, Rongtao Xu, Yan Yan, Xiaodan Liang, Ivan Laptev, and Xiaojun Chang. 2025. GLaD: Geometric Latent Distillation for Vision-Language-Action Models. *arXiv preprint arXiv:2512.09619* (2025). <https://arxiv.org/abs/2512.09619>
- [60] Yijing Guo, Mengjun Chao, Luo Wang, Tianyang Zhao, Haizhao Dai, Yingliang Zhang, Jingyi Yu, and Yujiao Shi. 2026. PanoVGGT: Feed-Forward 3D Reconstruction from Panoramic Imagery. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2603.17571>
- [61] Jisang Han, Sunghwan Hong, Jaewoo Jung, Wooseok Jang, Honggyu An, Qianqian Wang, Seungryong Kim, and Chen Feng. 2026. Emergent Outlier View Rejection in Visual Geometry Grounded Transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2512.04012>
- [62] Songjia He, Zixuan Chen, Hongyu Ding, Dian Shao, Jieqi Shi, Chenxu Li, Jing Huo, and Yang Gao. 2026. V-Dreamer: Automating Robotic Simulation and Trajectory Synthesis via Video Generation Priors. In *Workshop on Generative Digital Twins for Real2Sim and Sim2Real Transfer at the IEEE International Conference on Robotics and Automation (ICRA Workshop)*. <https://arxiv.org/abs/2603.18811>
- [63] Zhuolin He, Jing Li, Guanghao Li, Xiaolei Chen, Jiacheng Tang, Siyang Zhang, Zhounan Jin, Feipeng Cai, Bin Li, Jian Pu, Jia Cai, and Xiangyang Xue. 2026. DynamicVGGT: Learning Dynamic Point Maps for 4D Scene Reconstruction in Autonomous Driving. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 35670–35679. https://openaccess.thecvf.com/content/CVPR2026/html/He_DynamicVGGT_Learning_Dynamic_Point_Maps_for_4D_Scene_Reconstruction_in_CVPR_2026_paper.html
- [64] Peter Hedman, Julien Philip, True Price, Jan-Michael Frahm, George Drettakis, and Gabriel Brostow. 2018. Deep Blending for Free-Viewpoint Image-Based Rendering. *ACM Transactions on Graphics (TOG)* (2018). doi:10.1145/3272127.3275084
- [65] Javier Hidalgo-Carri6, Daniel Gehrig, and Davide Scaramuzza. 2020. Learning Monocular Dense Depth from Events. In *International Conference on 3D Vision (3DV)*. doi:10.1109/3dv50981.2020.00063
- [66] Mehdi Hosseinzadeh, Shin-Fang Chng, Yi Xu, Simon Lucey, Ian Reid, and Ravi Garg. 2025. G3Splat: Geometrically Consistent Generalizable Gaussian Splatting. *arXiv preprint arXiv:2512.17547* (2025). <https://arxiv.org/abs/2512.17547>
- [67] Lingxiang Hu, Naima Ait Oufroukh, Fabien Bonardi, and Raymond Ghandour. 2025. EC3R-SLAM: Efficient and Consistent Monocular Dense SLAM with Feed-Forward 3D Reconstruction. *arXiv preprint arXiv:2510.02080* (2025). <https://arxiv.org/abs/2510.02080>
- [68] Yu Hu, Chong Cheng, Sicheng Yu, Xiaoyang Guo, and Hao Wang. 2025. VGGT4D: Mining Motion Cues in Visual Geometry Transformers for 4D Scene Reconstruction. *arXiv preprint arXiv:2511.19971* (2025). <https://arxiv.org/abs/2511.19971>
- [69] David Huang, Guile Wu, Chengjie Huang, Bingbing Liu, and Dongfeng Bai. 2026. TurboVGGT: Fast Visual Geometry Reconstruction with Adaptive Alternating Attention. *arXiv preprint arXiv:2605.14315* (2026). <https://arxiv.org/abs/2605.14315>
- [70] Kaiyi Huang, Yukun Huang, Yu Li, Jianhong Bai, Xintao Wang, Zinan Lin, Xuefei Ning, Jiwen Yu, Pengfei Wan, Yu Wang, and Xihui Liu. 2026. CineScene: Implicit 3D as Effective Scene Representation for Cinematic Video Generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2602.06959>

- [71] Nan Huang, Pengcheng Yu, Weijia Zeng, James M. Rehg, Angjoo Kanazawa, Haiwen Feng, and Qianqian Wang. 2026. Self-Improving 4D Perception via Self-Distillation. *arXiv preprint arXiv:2604.08532* (2026). <https://arxiv.org/abs/2604.08532>
- [72] Ruopeng Huang, Boyu Yang, Wenlong Gui, Jeremy Morgan, Erdem Biyik, and Jiachen Li. 2026. SyncTwin: Fast Digital Twin Construction and Synchronization for Safe Robotic Manipulation. *arXiv preprint arXiv:2601.09920* (2026). <https://arxiv.org/abs/2601.09920>
- [73] Rasmus Ramsbøl Jensen, Anders Lindbjerg Dahl, George Vogiatzis, Engil Tola, and Henrik Aanæs. 2014. Large Scale Multi-view Stereopsis Evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2014.59
- [74] Jinwoo Jeon, Dong-Uk Seo, Eungchang Mason Lee, and Hyun Myung. 2026. AIM-SLAM: Dense Monocular SLAM via Adaptive and Informative Multi-View Keyframe Prioritization with Foundation Model. In *IEEE International Conference on Robotics and Automation (ICRA)*. <https://arxiv.org/abs/2603.05097>
- [75] Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, Dahua Lin, and Bo Dai. 2025. AnySplat: Feed-Forward 3D Gaussian Splatting from Unconstrained Views. *ACM Transactions on Graphics (TOG)* 44, 6 (2025), 1–16. doi:10.1145/3763326
- [76] Xianyun Jiao, Jingyi Xu, Zhongmiao Yan, Xieyuanli Chen, and Ling Pei. 2026. G2IA: Geometry-Guided Instance-Aware Retrieval and Refinement for Cross-Modal Place Recognition. *arXiv preprint arXiv:2606.15287* (2026). <https://arxiv.org/abs/2606.15287>
- [77] Wonseok Kang, Jaehyun Kim, Jeongmin Lee, and Tae-Wan Kim. 2026. Reference-Induced Consensus for Selective Posed-Reference Visual Localization. *arXiv preprint arXiv:2607.04722* (2026). <https://arxiv.org/abs/2607.04722>
- [78] Nikhil Keetha, Norman Müller, Johannes Schönberger, Lorenzo Porzi, Yuchen Zhang, Tobias Fischer, Arno Knapitsch, Duncan Zauss, Ethan Weber, Nelson Antunes, Jonathon Luiten, Manue Lopez-Antequera, Samuel Rota Bulò, Christian Richardt, Deva Ramanan, Sebastian Scherer, and Peter Kotschieder. 2026. MapAnything: Universal Feed-Forward Metric 3D Reconstruction. In *International Conference on 3D Vision (3DV)*. 499–509. doi:10.1109/3DV69130.2026.00054
- [79] Alex Kendall, Matthew Grimes, and Roberto Cipolla. 2015. PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. doi:10.1109/iccv.2015.336
- [80] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics (TOG)* 42, 4 (2023), 1–14. doi:10.1145/3592433
- [81] Alexander Khazatsky, Karl Pertsch, Suraj Nair, Ashwin Balakrishna, Sudeep Dasari, Siddharth Karamcheti, Soroush Nasiriany, Mohan Srirama, Lawrence Chen, Kirsty Ellis, Peter Fagan, Joey Hejna, Masha Itkina, Marion Lepert, Yecheng Ma, Patrick Miller, Jimmy Wu, Suneel Belkhale, Shivin Dass, Huy Ha, Arhan Jain, Abraham Lee, Youngwoon Lee, Marius Memmel, Sungjae Park, Ilija Radosavovic, Kaiyuan Wang, Albert Zhan, Kevin Black, Cheng Chi, Kyle Hatch, Shan Lin, Jingpei Lu, Jean Mercat, Abdul Rehman, Pannag Sanketi, Archit Sharma, Cody Simpson, Quan Vuong, Homer Walke, Blake Wulfe, Ted Xiao, Jonathan Yang, Arefeh Yavary, Tony Zhao, Christopher Agia, Rohan Bajjal, Mateo Castro, Daphne Chen, Qiuyu Chen, Trinity Chung, Jaimyn Drake, Ethan Foster, Jensen Gao, David Herrera, Minh Ho, Kyle Hsu, Jiaheng Hu, Donovan Jackson, Charlotte Le, Yunshuang Li, Roy Lin, Zehan Ma, Abhiram Maddukuri, Suvir Mirchandani, Daniel Morton, Tony Nguyen, Abigail O'Neill, Rosario Scalise, Derick Seale, Victor Son, Stephen Tian, Emi Tran, Andrew Wang, Yilin Wu, Annie Xie, Jingyun Yang, Patrick Yin, Yunchu Zhang, Osbert Bastani, Glen Berseth, Jeannette Bohg, Ken Goldberg, Abhinav Gupta, Abhishek Gupta, Dinesh Jayaraman, Joseph Lim, Jitendra Malik, Roberto Martín-Martín, Subramanian Ramamoorthy, Dorsa Sadigh, Shuran Song, Jiajun Wu, Michael Yip, Yuke Zhu, Thomas Kollar, Sergey Levine, and Chelsea Finn. 2024. DROID: A Large-Scale In-the-Wild Robot Manipulation Dataset. In *Robotics: Science and Systems (RSS)*. doi:10.15607/rss.2024.xx.120
- [82] Yongsung Kim, Wooseok Song, Jaihyun Lew, Hun Hwangbo, Jaehoon Lee, and Sungroh Yoon. 2026. HeSS: Head Sensitivity Score for Sparsity Redistribution in VGGT. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 36509–36517. https://openaccess.thecvf.com/content/CVPR2026/html/Kim_HeSS_Head_Sensitivity_Score_for_Sparsity_Redistribution_in_VGGT_CVPR_2026_paper.html
- [83] Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. 2017. Tanks and Temples: Benchmarking Large-Scale Scene Reconstruction. *ACM Transactions on Graphics (TOG)* (2017). doi:10.1145/3072959.3073599
- [84] Minseong Kweon, Wenyuan Zhao, Nuo Chen, Lulin Liu, Huiwen Han, Zihao Zhu, Srinivas Shakkottai, Chao Tian, and Zhiwen Fan. 2026. DarkVGGT: Seeing Through Darkness Using Thermal Geometry without Daylight Tax. *arXiv preprint arXiv:2606.11326* (2026). <https://arxiv.org/abs/2606.11326>
- [85] Yatong Lan, Rongkui Tang, and Lei He. 2026. Geo-EVS: Geometry-Conditioned Extrapolative View Synthesis for Autonomous Driving. *arXiv preprint arXiv:2604.07250* (2026). <https://arxiv.org/abs/2604.07250>
- [86] Alex Junho Lee, Younggun Cho, Young sik Shin, Ayoung Kim, and Hyun Myung. 2022. ViViD+: Vision for Visibility Dataset. *IEEE Robotics and Automation Letters (RA-L)* (2022). doi:10.1109/lra.2022.3168335
- [87] Jungho Lee, Minhyeok Lee, Sunghun Yang, Minseok Kang, and Sangyoun Lee. 2026. SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes. In *IEEE/CVF Conference on Computer Vision and Pattern*

- Recognition Findings (CVPR Findings). <https://Jho-Yonsei.github.io/SwiftVGGT/>
- [88] Seonho Lee, Jiho Choi, Inha Kang, Jiwook Kim, Junsung Park, and Hyunjung Shim. 2025. 3D-Aware Vision-Language Models Fine-Tuning with Geometric Distillation. In *Findings of the Association for Computational Linguistics: EMNLP (Findings of EMNLP)*. 10628–10647. doi:10.18653/v1/2025.findings-emnlp.562
- [89] Vincent Leroy, Yohann Cabon, and Jérôme Revaud. 2024. Grounding Image Matching in 3D with MAST3R. In *European Conference on Computer Vision (ECCV)*. 71–91. doi:10.1007/978-3-031-73220-1_5
- [90] Huan Li, Longjun Luo, Yuling Shi, and Xiaodong Gu. 2025. Analyzing the Mechanism of Attention Collapse in VGGT from a Dynamics Perspective. *arXiv preprint arXiv:2512.21691* (2025). <https://arxiv.org/abs/2512.21691>
- [91] Haotang Li, Zhenyu Qi, Shaohan Henry Wang, Kebin Peng, Zi Wang, Qing Guo, Sen He, and Huanrui Yang. 2026. PaceVGGT: Pre-Alternating-Attention Token Pruning for Visual Geometry Transformers. *arXiv preprint arXiv:2605.08371* (2026). <https://arxiv.org/abs/2605.08371>
- [92] Haoyuan Li, Wen Yang, Fang Xu, Hong Tan, Haijian Zhang, Shengyang Li, and Gui-Song Xia. 2026. Unifying UAV Cross-View Geo-Localization via 3D Geometric Perception. *arXiv preprint arXiv:2604.01747* (2026). <https://arxiv.org/abs/2604.01747>
- [93] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Karl Pertsch, Oier Mees, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. 2024. Evaluating Real-World Robot Manipulation Policies in Simulation. In *Conference on Robot Learning (CoRL)*. <https://arxiv.org/abs/2405.05941>
- [94] Zhengqi Li and Noah Snavely. 2018. MegaDepth: Learning Single-View Depth Prediction from Internet Photos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2018.00218
- [95] Yiyi Liao, Jun Xie, and Andreas Geiger. 2022. KITTI-360: A Novel Dataset and Benchmarks for Urban Scene Understanding in 2D and 3D. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* (2022). doi:10.1109/tpami.2022.3179507
- [96] Junhong Lin, Kangli Wang, Shunzhou Wang, Songlin Fan, Ge Li, and Wei Gao. 2025. VGD: Visual Geometry Gaussian Splatting for Feed-Forward Surround-View Driving Reconstruction. *arXiv preprint arXiv:2510.19578* (2025). <https://arxiv.org/abs/2510.19578>
- [97] Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, Xuanmao Li, Xingpeng Sun, Rohan Ashok, Aniruddha Mukherjee, Hao Kang, Xiangrui Kong, Gang Hua, Tianyi Zhang, Bedrich Benes, and Aniket Bera. 2024. DL3DV-10K: A Large-Scale Scene Dataset for Deep Learning-Based 3D Vision. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr52733.2024.02092
- [98] Andrew Liu, Richard Tucker, Varun Jampani, Ameesh Makadia, Noah Snavely, and Angjoo Kanazawa. 2021. Infinite Nature: Perpetual View Generation of Natural Scenes from a Single Image. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. doi:10.1109/iccv48922.2021.01419
- [99] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. 2023. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. 44776–44791. doi:10.52202/075280-1939
- [100] Xuanyi Liu, Deyi Ji, Liqun Liu, Lanyun Zhu, Xuhang Chen, Qianxiong Xu, Peng Shu, Huan Yu, Jie Jiang, Feng Gao, and Siwei Ma. 2026. CamGeo: Sparse Camera-Conditioned Image-to-Video Generation with 3D Geometry Priors. In *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2605.30895>
- [101] Xuanyi Liu, Chunan Yu, Deyi Ji, Qi Zhu, Lingyun Sun, Xuanfu Li, Jin Ma, Tianrun Chen, and Lanyun Zhu. 2026. StreamCacheVGGT: Streaming Visual Geometry Transformers with Robust Scoring and Hybrid Cache Compression. *arXiv preprint arXiv:2604.15237* (2026). <https://arxiv.org/abs/2604.15237>
- [102] Yunze Liu, Yun Liu, Che Jiang, Kangbo Lyu, Weikang Wan, Hao Shen, Boqiang Liang, Zhoujie Fu, He Wang, and Li Yi. 2022. HOI4D: A 4D Egocentric Dataset for Category-Level Human-Object Interaction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 21013–21022. https://openaccess.thecvf.com/content/CVPR2022/html/Liu_HOI4D_A_4D_Egocentric_Dataset_for_Category-Level_Human-Object_Interaction_CVPR_2022_paper.html
- [103] Yang Liu, Chuanchen Luo, Zimo Tang, Junran Peng, and Zhaoxiang Zhang. 2025. VGGT-X: When VGGT Meets Dense Novel View Synthesis. *arXiv preprint arXiv:2509.25191* (2025). <https://arxiv.org/abs/2509.25191>
- [104] Yifan Liu, Zhiyuan Min, Zhenwei Wang, Junta Wu, Tengfei Wang, Yixuan Yuan, Yawei Luo, and Chunchao Guo. 2026. WorldMirror: Universal 3D World Reconstruction with Any-Prior Prompting. In *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2510.10726>
- [105] Yifan Liu, Bin Peng, Mingke Xing, Tong Hua, and Jincai Chen. 2026. Water-VGGT: Leveraging Visual Geometry and Water-Optics for Robust 3D Reconstruction in Underwater Environments. *Ocean Engineering* 364 (2026), 127044. doi:10.1016/j.oceaneng.2026.127044
- [106] Jiahui Lu, Haihong Xiao, Xueyan Zhao, and Wenxiong Kang. 2025. Gesplat: Robust Pose-Free 3D Reconstruction via Geometry-Guided Gaussian Splatting. *arXiv preprint arXiv:2510.10097* (2025). <https://arxiv.org/abs/2510.10097>

- [107] Si-Yu Lu, Po-Ting Chen, Hui-Che Hsu, Sin-Ye Jhong, Wen-Huang Cheng, and Yung-Yao Chen. 2026. OVGGT: O(1) Constant-Cost Streaming Visual Geometry Transformer. In *European Conference on Computer Vision (ECCV)*. <https://vairsr.github.io/OVGGT/>
- [108] Will Maddern, Geoffrey Pascoe, Chris Linegar, and Paul Newman. 2017. 1 Year, 1000 km: The Oxford RobotCar Dataset. *The International Journal of Robotics Research (IJRR)* (2017). doi:10.1177/0278364916679498
- [109] Dominic Maggio and Luca Carlone. 2026. VGGT-SLAM 2.0: Real-time Dense Feed-forward Scene Reconstruction. In *Robotics: Science and Systems (RSS)*. <https://arxiv.org/abs/2601.19887>
- [110] Dominic Maggio, Hyungtae Lim, and Luca Carlone. 2025. VGGT-SLAM: Dense RGB SLAM Optimized on the SL(4) Manifold. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 38. <https://arxiv.org/abs/2505.12549>
- [111] Soroush Mahdi, Fardin Ayar, Ehsan Javanmardi, Manabu Tsukada, and Mahdi Javanmardi. 2025. Evict3R: Training-Free Token Eviction for Memory-Bounded Streaming Visual Geometry Transformers. *arXiv preprint arXiv:2509.17650* (2025). <https://arxiv.org/abs/2509.17650>
- [112] Avilasha Mandal, Rajesh Kumar, Sudarshan Sunil Harithas, and Chetan Arora. 2026. VGGT-SLAM++. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops)*. <https://arxiv.org/abs/2604.06830>
- [113] Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. 2022. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *IEEE Robotics and Automation Letters (RA-L)* (2022). doi:10.1109/ra.2022.3180108
- [114] Ben Mildenhall, Pratul P. Srinivasan, Rodrigo Ortiz-Cayon, Nima Khademi Kalantari, Ravi Ramamoorthi, Ren Ng, and Abhishek Kar. 2019. Local Light Field Fusion: Practical View Synthesis with Prescriptive Sampling Guidelines. *ACM Transactions on Graphics (TOG)* (2019). doi:10.1145/3306346.3322980
- [115] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *European Conference on Computer Vision (ECCV)*. 405–421. doi:10.1007/978-3-030-58452-8_24
- [116] Juhong Min, Jongmin Lee, Jean Ponce, and Minsu Cho. 2019. SPair-71k: A Large-Scale Benchmark for Semantic Correspondence. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. doi:10.1109/ICCV.2019.00372
- [117] Zehao Ni, Yonghao He, Lingfeng Qian, Jilei Mao, Fa Fu, Wei Sui, Hu Su, Junran Peng, Zhipeng Wang, and Bin He. 2025. VO-DP: Semantic-Geometric Adaptive Diffusion Policy for Vision-Only Robotic Manipulation. *arXiv preprint arXiv:2510.15530* (2025). <https://arxiv.org/abs/2510.15530>
- [118] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mido Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. 2024. DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research (TMLR)* (2024). <https://openreview.net/forum?id=a68SUt6zFt>
- [119] Emanuele Palazzolo, Jens Behley, Philipp Lottes, Philippe Giguère, and Cyrill Stachniss. 2019. ReFusion: 3D Reconstruction in Dynamic Environments for RGB-D Cameras Exploiting Residuals. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. doi:10.1109/iros40897.2019.8967590
- [120] Junxiang Pan, Lipu Zhou, and Baojie Chen. 2026. HyVGGT-VO: Tightly Coupled Hybrid Dense Visual Odometry with Feed-Forward Models. *arXiv preprint arXiv:2604.02107* (2026). <https://arxiv.org/abs/2604.02107>
- [121] Sicheng Pan, Chen Tang, Shuzhao Xie, Ke Yang, Weixiang Zhang, Jiawei Li, Bin Chen, Shu-Tao Xia, and Zhi Wang. 2026. Tail-Aware Post-Training Quantization for 3D Geometry Models. *arXiv preprint arXiv:2602.01741* (2026). <https://arxiv.org/abs/2602.01741>
- [122] Tianbo Pan, Xingyi Yang, Shizun Wang, and Xinchao Wang. 2026. Visual Geometry Transformer in the Wild: Distractor-Free 3D Reconstruction. *arXiv preprint arXiv:2606.22787* (2026). <https://arxiv.org/abs/2606.22787>
- [123] Xiaqing Pan, Nicholas Charron, Yongqian Yang, Scott Peters, Thomas Whelan, Chen Kong, Omkar Parkhi, Richard Newcombe, and Yuheng Carl Ren. 2023. Aria Digital Twin: A New Benchmark Dataset for Egocentric 3D Machine Perception. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. doi:10.1109/iccv51070.2023.01842
- [124] Zhizhen Pan, Hesong Wang, and Huan Wang. 2026. QVGGT: Post-Training Quantized Visual Geometry Grounded Transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2605.31124>
- [125] Jinsoo Park, Donggyu Choi, Ahyun Seo, Minsu Cho, and Jeany Son. 2026. Diversity-Aware View Partitioning for Scalable VGGT. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2607.01885>
- [126] Keunhong Park, Utkarsh Sinha, Jonathan T. Barron, Sofien Bouaziz, Dan B. Goldman, Steven M. Seitz, and Ricardo Martin-Brualla. 2021. HyperNeRF: A Higher-Dimensional Representation for Topologically Varying Neural Radiance Fields. *ACM Transactions on Graphics (TOG)* (2021). doi:10.1145/3450626.3459882
- [127] Haosong Peng, Hao Li, Yalun Dai, Yushi Lan, Yihang Luo, Tianyu Qi, Zhengshen Zhang, Yufeng Zhan, Junfei Zhang, Wenchao Xu, and Ziwei Liu. 2026. OmniVGGT: Omni-Modality Driven Visual Geometry

- Grounded Transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 36485–36497. https://openaccess.thecvf.com/content/CVPR2026/html/Peng_OmniVGGT_Omni-Modality_Driven_Visual_Geometry_Grounded_Transformer_CVPR_2026_paper.html
- [128] Federico Perazzi, Jordi Pont-Tuset, Brian McWilliams, Luc Van Gool, Markus Gross, and Alexander Sorkine-Hornung. 2016. A Benchmark Dataset and Evaluation Methodology for Video Object Segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2016.85
- [129] Zezhong Qian, Xiaowei Chi, Yuming Li, Shizun Wang, Zhiyuan Qin, Xiaozhu Ju, Sirui Han, and Shanghang Zhang. 2025. WristWorld: Generating Wrist-Views via 4D World Models for Robotic Manipulation. *arXiv preprint arXiv:2510.07313* (2025). <https://arxiv.org/abs/2510.07313>
- [130] Kai Qin, Jing Li, Sisi Zlatanova, Haitao Wu, Hao Wu, Yin Gao, Dingjie Zhou, Yuchen Li, Sizhe Shen, Xiangjun Qu, Zhenxin Zhang, Banghui Yang, and Shicheng Xu. 2026. VGGT-Geo: Probabilistic Geometric Fusion of Visual Geometry Grounded Transformer Priors for Robust Dense Indoor SLAM. *ISPRS International Journal of Geo-Information (IJGI)* 15, 2 (2026), 85. doi:10.3390/ijgi15020085
- [131] Jinyuan Qu, Hongyang Li, and Lei Zhang. 2026. SegVGGT: Joint 3D Reconstruction and Instance Segmentation from Multi-View Images. In *European Conference on Computer Vision (ECCV)*. <https://github.com/IDEA-Research/SegVGGT>
- [132] Zhifeng Rao, Wenlong Chen, Lei Xie, Xia Hua, Dongfu Yin, Zhen Tian, and F. Richard Yu. 2026. AugVLA-3D: Depth-Driven Feature Augmentation for Vision-Language-Action Models. *arXiv preprint arXiv:2602.10698* (2026). <https://arxiv.org/abs/2602.10698>
- [133] Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. 2021. Common Objects in 3D: Large-Scale Learning and Evaluation of Real-Life 3D Category Reconstruction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. doi:10.1109/iccv48922.2021.01072
- [134] Jiangwei Ren, Xingyu Jiang, Zijie Song, Wei Xu, Hongkai Lin, Dingkan Liang, and Xiang Bai. 2026. Wat3R: Underwater 3D Geometry Learning without Annotations. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2607.08772>
- [135] Yinrui Ren, Jinjing Zhu, Kanghao Chen, Zhuoxiao Li, Jing Ou, Zidong Cao, Tongyan Hua, Peilun Shi, Yingchun Fu, Wufan Zhao, and Hui Xiong. 2026. EventVGGT: Exploring Cross-Modal Distillation for Consistent Event-based Depth Estimation. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2603.09385>
- [136] Manuel Rey-Area, Mingze Yuan, and Christian Richardt. 2022. 360MonoDepth: High-Resolution 360deg Monocular Depth Estimation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 3762–3772. doi:10.1109/CVPR52688.2022.00374
- [137] David Rozenberszki, Or Litany, and Angela Dai. 2022. Language-Grounded Indoor 3D Semantic Segmentation in the Wild. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2204.07761>
- [138] Paul-Edouard Sarlin, Mihai Dusmanu, Johannes L. Schönberger, Victor Larsson, and Marc Pollefeys. 2022. LaMAR: Benchmarking Localization and Mapping for Augmented Reality. In *European Conference on Computer Vision (ECCV)*. doi:10.1007/978-3-031-20071-7_40
- [139] Johannes L. Schönberger and Jan-Michael Frahm. 2016. Structure-from-Motion Revisited. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 4104–4113. doi:10.1109/CVPR.2016.445
- [140] Johannes L. Schönberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. 2016. Pixelwise View Selection for Unstructured Multi-View Stereo. In *European Conference on Computer Vision (ECCV)*. 501–518. doi:10.1007/978-3-319-46487-9_31
- [141] Thomas Schöps, Johannes L. Schönberger, Silvano Galliani, Torsten Sattler, Konrad Schindler, Marc Pollefeys, and Andreas Geiger. 2017. A Multi-View Stereo Benchmark with High-Resolution Images and Multi-Camera Videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2017.272
- [142] David Schubert, Thore Goll, Nikolaus Demmel, Vladyslav Usenko, Jörg Stückler, and Daniel Cremers. 2018. The TUM VI Benchmark for Evaluating Visual-Inertial Odometry. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 1680–1687. doi:10.1109/IROS.2018.8593419
- [143] You Shen, Zhipeng Zhang, Yansong Qu, Xiawu Zheng, Jiayi Ji, Shengchuan Zhang, and Liujuan Cao. 2026. FastVGGT: Training-Free Acceleration of Visual Geometry Transformer. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2509.02560>
- [144] Jamie Shotton, Ben Glocker, Christopher Zach, Shahram Izadi, Antonio Criminisi, and Andrew Fitzgibbon. 2013. Scene Coordinate Regression Forests for Camera Relocalization in RGB-D Images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr.2013.377
- [145] Zhijian Shu, Cheng Lin, Tao Xie, Wei Yin, Ben Li, Zhiyuan Pu, Weize Li, Yao Yao, Xun Cao, Xiaoyang Guo, and Xiao-Xiao Long. 2026. LiteVGGT: Boosting Vanilla VGGT via Geometry-Aware Cached Token Merging. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 36422–36432. https://openaccess.thecvf.com/content/CVPR2026/html/Shu_LiteVGGT_Boosting_Vanilla_VGGT_via_Geometry-aware_Cached-Token_Merging_CVPR_2026_paper.html

- [146] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. 2012. Indoor Segmentation and Support Inference from RGBD Images. In *European Conference on Computer Vision (ECCV)*. doi:10.1007/978-3-642-33715-4_54
- [147] Daniil Sinitsyn, Nikita Arslanov, and Daniel Cremers. 2026. RayTun3R: Online Camera Adaptation in 3D Foundation Models. *arXiv preprint arXiv:2607.02711* (2026). <https://arxiv.org/abs/2607.02711>
- [148] Julian Straub, Thomas Whelan, Lingni Ma, Yufan Chen, Erik Wijmans, Simon Green, Jakob J. Engel, Raul Mur-Artal, Carl Ren, Shobhit Verma, Anton Clarkson, Mingfei Yan, Brian Budge, Yajie Yan, Xiaqing Pan, June Yon, Yuyang Zou, Kimberly Leon, Nigel Carter, Jesus Briales, Tyler Gillingham, Elias Mueggler, Luis Pesqueira, Manolis Savva, Dhruv Batra, Hauke M. Strasdat, Renzo De Nardi, Michael Goesele, Steven Lovegrove, and Richard Newcombe. 2019. The Replica Dataset: A Digital Replica of Indoor Spaces. *arXiv preprint arXiv:1906.05797* (2019). <https://arxiv.org/abs/1906.05797>
- [149] Jürgen Sturm, Nikolaus Engelhard, Felix Endres, Wolfram Burgard, and Daniel Cremers. 2012. A Benchmark for the Evaluation of RGB-D SLAM Systems. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. doi:10.1109/iros.2012.6385773
- [150] Zunhai Su, Weihao Ye, Hansen Feng, Keyu Fan, Jing Zhang, Dahai Yu, Zhengwu Liu, and Ngai Wong. 2026. XStreamVGGT: Extremely Memory-Efficient Streaming Vision Geometry Grounded Transformer with KV Cache Compression. In *Society for Information Display Display Week (SID Display Week)*. <https://arxiv.org/abs/2601.01204>
- [151] Edgar Sucar, Eldar Insafutdinov, Zihang Lai, and Andrea Vedaldi. 2026. V-DPM: 4D Video Reconstruction with Dynamic Point Maps. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2601.09499>
- [152] Pei Sun, Henrik Kretschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James Guo, Yin Zhou, Yuning Chai, Benjamin Caine, Vijay Vasudevan, Wei Han, Jiquan Ngiam, Hang Zhao, Aleksei Timofeev, Scott Ettinger, Maxim Krivokon, Amy Gao, Aditya Joshi, Yu Zhang, Jonathon Shlens, Zhifeng Chen, and Dragomir Anguelov. 2020. Scalability in Perception for Autonomous Driving: Waymo Open Dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr42600.2020.00252
- [153] Xiangyu Sun, Shijie Wang, Fengyi Zhang, Lin Liu, Caiyan Jia, Ziyang Song, Zi Huang, and Yadan Luo. 2026. VGGT-World: Transforming VGGT into an Autoregressive Geometry World Model. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2603.12655>
- [154] Xianbing Sun, Zhikai Zhu, Zhengyu Lou, Bo Yang, Jinyang Tang, Liqing Zhang, He Wang, and Jianfu Zhang. 2026. AVGGT: Rethinking Global Attention for Accelerating VGGT. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 251–260. https://openaccess.thecvf.com/content/CVPR2026/html/Sun_AVGGT_Rethinking_Global_Attention_for_Accelerating_VGGT_CVPR_2026_paper.html
- [155] Niko Sünderhauf, Peer Neubert, and Peter Protzel. 2013. Are We There Yet? Challenging SeqSLAM on a 3000 km Journey Across All Four Seasons. In *Workshop on Long-Term Autonomy at the IEEE International Conference on Robotics and Automation (ICRA Workshop)*. <https://dblp.org/rec/conf/icra/SunderhaufNP13.html>
- [156] Shaofeng Tan, Xiaoguang Ma, and Xizi Jia. 2025. VGGT-Mapping: A Mapping System for Robot Navigation based on Visual Geometry Grounded Transformer. In *International Conference on Intelligent Robotics and Automatic Control*. 160–165. doi:10.1109/IRAC67707.2025.11381128
- [157] Zecheng Tang, Jiaye Fu, Qiankun Gao, Haijie Li, Yanmin Wu, Jiaqi Zhang, Siwei Ma, and Jian Zhang. 2026. Spark3R: Asymmetric Token Reduction Makes Fast Feed-Forward 3D Reconstruction. *arXiv preprint arXiv:2605.06270* (2026). <https://arxiv.org/abs/2605.06270>
- [158] Peilin Tao, Chong Cheng, Yuanshen Du, Caiwei Song, Zhengqing Chen, Xiaoyang Guo, Wei Yin, Weiqiang Ren, Qian Zhang, Hainan Cui, and Shuhan Shen. 2026. Anchor3R: Streaming 3D Reconstruction with Transient Anchors for Long-Horizon Visual Mapping. *arXiv preprint arXiv:2606.05035* (2026). <https://arxiv.org/abs/2606.05035>
- [159] Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse kai Chan, Yuan Gao, Xuanlin Li, Tongzhou Mu, Nan Xiao, Arnav Gurha, Viswesh Nagaswamy Rajesh, Yong Woo Choi, Yen-Ru Chen, Zhiao Huang, Roberto Calandra, Rui Chen, Shan Luo, and Hao Su. 2025. ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI. In *Robotics: Science and Systems (RSS)*. <https://arxiv.org/abs/2410.00425>
- [160] Yifu Tao, Miguel Ángel Muñoz-Bañón, Lintong Zhang, Jiahao Wang, Lanke Frank Tarimo Fu, and Maurice Fallon. 2025. The Oxford Spires Dataset: Benchmarking Large-Scale LiDAR-Visual Localisation, Reconstruction and Radiance Field Methods. *The International Journal of Robotics Research (IJRR)* (2025). <https://arxiv.org/abs/2411.10546>
- [161] Joachim Tesch, Georgios Becherini, Pramod Achar, Anastasios Yiannakidis, Muhammed Kocabas, Partha Patel, and Michael J. Black. 2025. BEDLAM2.0: Synthetic Humans and Cameras in Motion. In *Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS Datasets and Benchmarks)*.
- [162] An Dinh Vuong, Minh Nhat Vu, and Ian Reid. 2025. Improving Robotic Manipulation with Efficient Geometry-Aware Vision Encoder. *arXiv preprint arXiv:2509.15880* (2025). <https://arxiv.org/abs/2509.15880>

- [163] Yecong Wan, Fan Li, Mingwen Shao, and Wangmeng Zuo. 2026. SplatWeaver: Learning to Allocate Gaussian Primitives for Generalizable Novel View Synthesis. *arXiv preprint arXiv:2605.07287* (2026). <https://arxiv.org/abs/2605.07287>
- [164] Bo Wang, Jiehong Lin, Chenzhi Liu, Xinting Hu, Yifei Yu, Tianjia Liu, Zhongrui Wang, and Xiaojuan Qi. 2025. MG-Nav: Dual-Scale Visual Navigation via Sparse Spatial Memory. *arXiv preprint arXiv:2511.22609* (2025). <https://arxiv.org/abs/2511.22609>
- [165] Chung-Shien Brian Wang, Christian Schmidt, Jens Piekenbrinck, and Bastian Leibe. 2026. Block-Sparse Global Attention for Efficient Multi-View Geometry Transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14546–14555. https://openaccess.thecvf.com/content/CVPR2026/html/Wang_Block-Sparse_Global_Attention_for_Efficient_Multi-View_Geometry_Transformers_CVPR_2026_paper.html
- [166] Haonan Wang, Hanyu Zhou, Haoyue Liu, and Luxin Yan. 2026. 4D-VGGT: A SpatioTemporal Foundation Model for Dynamic Scene Geometry Estimation. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2511.18416>
- [167] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. 2025. VGGT: Visual Geometry Grounded Transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr52734.2025.00499
- [168] Jianyuan Wang, Minghao Chen, Shangzhan Zhang, Nikita Karaev, Johannes Schönberger, Patrick Labatut, Piotr Bojanowski, David Novotny, Andrea Vedaldi, and Christian Rupprecht. 2026. VGGT-Ω. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 21486–21499. https://openaccess.thecvf.com/content/CVPR2026/html/Wang_VGGT-ohm_CVPR_2026_paper.html
- [169] Jie Wang, Guang Li, Zhijian Huang, Chenxu Dang, Hangjun Ye, Yahong Han, and Long Chen. 2026. VGGDrive: Empowering Vision-Language Models with Cross-View Geometric Grounding for Autonomous Driving. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10954–10964. https://openaccess.thecvf.com/content/CVPR2026/html/Wang_VGGDrive_Empowering_Vision-Language_Models_with_Cross-View_Geometric_Grounding_for_Autonomous_CVPR_2026_paper.html
- [170] Kuanning Wang, Ke Fan, Yuqian Fu, Siyu Lin, Hu Luo, Daniel Seita, Yanwei Fu, Yu-Gang Jiang, and Xiangyang Xue. 2026. OCRA: Object-Centric Learning with 3D and Tactile Priors for Human-to-Robot Action Transfer. In *IEEE International Conference on Robotics and Automation (ICRA)*. <https://arxiv.org/abs/2603.14401>
- [171] Lijie Wang, Lianjie Guo, Ziyi Xu, Qianhao Wang, Fei Gao, and Xieyuanli Chen. 2026. LiDAR-VGGT: Cross-Modal Coarse-to-Fine Fusion for Globally Consistent and Metric-Scale Dense Mapping. *IEEE Robotics and Automation Letters (RA-L)* 11, 4 (2026), 4721–4728. doi:10.1109/LRA.2026.3666387
- [172] Qianqian Wang, Yifei Zhang, Aleksander Holynski, Alexei A. Efros, and Angjoo Kanazawa. 2025. Continuous 3D Perception Model with Persistent State. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10510–10522. doi:10.1109/CVPR52734.2025.00983
- [173] Runze Wang, Yuxuan Song, Youcheng Cai, and Ligang Liu. 2026. STAC: Plug-and-Play Spatio-Temporal Aware Cache Compression for Streaming 3D Reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 7567–7576. https://openaccess.thecvf.com/content/CVPR2026/html/Wang_STAC_Plug-and-Play_Spatio-Temporal_Aware_Cache_Compression_for_Streaming_3D_Reconstruction_CVPR_2026_paper.html
- [174] Shenglin Wang, Mingtong Dai, Jingxuan Su, Lingbo Liu, Chunjie Chen, Xinyu Wu, and Liang Lin. 2025. GraspView: Active Perception Scoring and Best-View Optimization for Robotic Grasping in Cluttered Environments. *arXiv preprint arXiv:2511.04199* (2025). <https://arxiv.org/abs/2511.04199>
- [175] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jérôme Revaud. 2024. DUST3R: Geometric 3D Vision Made Easy. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. doi:10.1109/cvpr52733.2024.01956
- [176] Weitian Wang, Lukas Meiner, Rai Shubham, Cecilia De La Parra, and Akash Kumar. 2026. HTTPM: Head-wise Temporal Token Merging for Faster VGGT. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 26379–26388. https://openaccess.thecvf.com/content/CVPR2026/html/Wang_HTTPM_Head-wise_Temporal-Token_Merging_for_Faster_VGGT_CVPR_2026_paper.html
- [177] Wenshan Wang, Delong Zhu, Xiangwei Wang, Yaoyu Hu, Yuheng Qiu, Chen Wang, Yafei Hu, Ashish Kapoor, and Sebastian Scherer. 2020. TartanAir: A Dataset to Push the Limits of Visual SLAM. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. doi:10.1109/iros45743.2020.9341801
- [178] Zipeng Wang and Dan Xu. 2026. FlashVGGT: Efficient and Scalable Visual Geometry Transformers with Compressed Descriptor Attention. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 21826–21835. https://openaccess.thecvf.com/content/CVPR2026/html/Wang_FlashVGGT_Efficient_and_Scalable_Visual_Geometry_Transformers_with_Compressed_Descriptor_CVPR_2026_paper.html

- [179] Changli Wu, Haodong Wang, Jiayi Ji, Yutian Yao, Chunsai Du, Jihua Kang, Yanwei Fu, and Liujuan Cao. 2026. MVGGT: Multimodal Visual Geometry Grounded Transformer for Multiview 3D Referring Expression Segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 16800–16810. https://openaccess.thecvf.com/content/CVPR2026/html/Wu_MVGGT_Multimodal_Visual_Geometry_Grounded_Transformer_for_Multiview_3D_Referring_CVPR_2026_paper.html
- [180] Xianfeng Wu, Yajing Bai, Minghan Li, Xianzu Wu, Xueqi Zhao, Zhongyuan Lai, Wenyu Liu, and Xinggang Wang. 2025. 4DLangVGGT: 4D Language-Visual Geometry Grounded Transformer. *arXiv preprint arXiv:2512.05060* (2025). <https://arxiv.org/abs/2512.05060>
- [181] Peng Xiang, Liang Han, Hui Zhang, Yu-Shen Liu, and Zhizhong Han. 2026. VGGs: VGGT-guided Gaussian Splatting for Efficient and Faithful Sparse-View Surface Reconstruction. In *AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 40. 10969–10977. doi:10.1609/aaai.v40i13.38074
- [182] Wenhui Xiao, Remi Chierchia, Rodrigo Santa Cruz, Xuesong Li, David Ahmedt-Aristizabal, Olivier Salvado, Clinton Fookes, and Leo Lebrat. 2025. Neural Radiance Fields for the Real World: A Survey. *ACM Computing Surveys (CSUR)* (2025). <https://arxiv.org/abs/2501.13104>
- [183] Zhuang Xiong, Chen Zhang, Qingshan Xu, and Wenbing Tao. 2026. VGGT-Motion: Motion-Aware Calibration-Free Monocular SLAM for Long-Range Consistency. In *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2602.05508>
- [184] Jingyi Xu, Zhangshuo Qi, Zhongmiao Yan, Xuyu Gao, Qianyun Jiao, Songpengcheng Xia, Xieyuanli Chen, and Ling Pei. 2026. VGGT-MPR: VGGT-Enhanced Multimodal Place Recognition in Autonomous Driving Environments. *arXiv preprint arXiv:2602.19735* (2026). <https://arxiv.org/abs/2602.19735>
- [185] Yangfan Xu, Lilian Zhang, Xiaofeng He, Pengdong Wu, Wenqi Wu, and Jun Mao. 2026. GPA-VGGT: Adapting VGGT to Large Scale Localization by Self-Supervised Learning with Geometry and Physics Aware Loss. *arXiv preprint arXiv:2601.16885* (2026). <https://arxiv.org/abs/2601.16885>
- [186] Zhisong Xu and Takeshi Oishi. 2026. FrameVGGT: Coherence-Preserving Memory for Bounded Streaming Geometry. *arXiv preprint arXiv:2603.07690* (2026). <https://arxiv.org/abs/2603.07690>
- [187] Zhisong Xu and Takeshi Oishi. 2026. PanolMager: Geometry-Guided Novel View Synthesis and Reconstruction from Sparse Panoramic Views. *arXiv preprint arXiv:2606.27071* (2026). <https://arxiv.org/abs/2606.27071>
- [188] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, and Matt Feiszli. 2025. Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 21924–21935. doi:10.1109/CVPR52734.2025.02042
- [189] Songlin Yang, Tianyi Wei, Yushi Lan, Zeqi Xiao, Anyi Rao, and Xingang Pan. 2026. Dense Semantic Matching with VGGT Prior. In *European Conference on Computer Vision (ECCV)*. <https://arxiv.org/abs/2509.21263>
- [190] Yurou Yang, Muyuan Lin, Roberto Martin-Martin, Martin Labrie, Shreekanth Gayaka, Cheng-Hao Kuo, and Luca Carlone. 2026. Understanding the Impact of Geometric Foundation Models on Vision-Language-Action Models. *arXiv preprint arXiv:2605.24642* (2026). <https://arxiv.org/abs/2605.24642>
- [191] Yandan Yang, Shuang Zeng, Tong Lin, Xinyuan Chang, Dekang Qi, Junjin Xiao, Haoyun Liu, Ronghan Chen, Yuzhi Chen, Dongjie Huo, Feng Xiong, Xing Wei, Zhiheng Ma, and Mu Xu. 2026. ABot-M0: VLA Foundation Model for Robotic Manipulation with Action Manifold Learning. *arXiv preprint arXiv:2602.11236* (2026). <https://arxiv.org/abs/2602.11236>
- [192] Chandan Yeshwanth, Yueh-Cheng Liu, Matthias Nießner, and Angela Dai. 2023. ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 12–22. doi:10.1109/ICCV51070.2023.00008
- [193] Jinhao You, Shuo Lyu, Zhuohang Lyu, Tanxuan Li, Zibo Zhao, Jiaxiang Hu, Kai Tang, and Yichen Guo. 2026. RegimeVGGT: Layer-Wise Spatially Preserving Redundancy Removal for Visual Geometry Grounded Transformer. *arXiv preprint arXiv:2606.18439* (2026). <https://arxiv.org/abs/2606.18439>
- [194] Bin Yu, Shijie Lian, Xiaopeng Lin, Zhaolong Shen, Yuliang Wei, Haishan Liu, Changti Wu, Hang Yuan, Bailing Wang, Cong Huang, and Kai Chen. 2026. 3D-Mix for VLA: A Plug-and-Play Module for Integrating VGGT-Based 3D Information into Vision-Language-Action Models. *arXiv preprint arXiv:2603.24393* (2026). <https://arxiv.org/abs/2603.24393>
- [195] Haibao Yu, Kuntao Xiao, Jiahang Wang, Ruiyang Hao, Yuxin Huang, Guoran Hu, Haifang Qin, Bowen Jing, Yuntian Bo, and Ping Luo. 2026. ReconDrive: Fast Feed-Forward 4D Gaussian Splatting for Autonomous Driving Scene Reconstruction. *arXiv preprint arXiv:2603.07552* (2026). <https://arxiv.org/abs/2603.07552>
- [196] Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. 2020. Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning. In *Proceedings of the Conference on Robot Learning (CoRL)*. <https://proceedings.mlr.press/v100/yu20a.html>

- [197] Jiayi Yuan, Haobo Jiang, De Wen Soh, and Na Zhao. 2026. VGGT-360: Geometry-Consistent Zero-Shot Panoramic Depth Estimation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 19874–19883. https://openaccess.thecvf.com/content/CVPR2026/html/Yuan_VGGT-360_Geometry-Consistent_Zero-Shot_Panoramic_Depth_Estimation_CVPR_2026_paper.html
- [198] Shuai Yuan, Yantai Yang, Xiaotian Yang, Xupeng Zhang, Zhonghao Zhao, Lingming Zhang, and Zhipeng Zhang. 2026. InfiniteVGGT: Visual Geometry Grounded Transformer for Endless Streams. *arXiv preprint arXiv:2601.02281* (2026). <https://arxiv.org/abs/2601.02281>
- [199] Sungjae Yun, Minkyu Jung, Jisung Kim, Seungwon Jung, Younggun Cho, Myeongjin Jeon, Gunhee Kim, and Ayoung Kim. 2022. STheReO: Stereo Thermal Dataset for Research in Odometry and Mapping. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. doi:10.1109/iroso47612.2022.9981857
- [200] Ying Zang, Yidong Han, Chaotao Ding, Yuanqi Hu, Deyi Ji, Qi Zhu, Xuanfu Li, Jin Ma, Lingyun Sun, Tianrun Chen, and Lanyun Zhu. 2026. Robust 4D Visual Geometry Transformer with Uncertainty-Aware Priors. *arXiv preprint arXiv:2604.09366* (2026). <https://arxiv.org/abs/2604.09366>
- [201] Ying Zang, Xuanyi Liu, Yidong Han, Deyi Ji, Chaotao Ding, Yuanqi Hu, Qi Zhu, Xuanfu Li, Jin Ma, Lingyun Sun, Tianrun Chen, and Lanyun Zhu. 2026. 4DVGGT-D: 4D Visual Geometry Transformer with Improved Dynamic Depth Estimation. *arXiv preprint arXiv:2605.12027* (2026). <https://arxiv.org/abs/2605.12027>
- [202] Hongyang Zhang, Yin hao Liu, Haitao Zhang, Zhongyi Wen, Zhenyu Kuang, Shuxian Liang, and Xiansheng Hua. 2026. GeoLink: A 3D-Aware Framework Towards Better Generalization in Cross-View Geo-Localization. *arXiv preprint arXiv:2604.13183* (2026). <https://arxiv.org/abs/2604.13183>
- [203] Haoyu Zhang, Zeyu Zhang, Zedong Zhou, Yang Zhao, and Hao Tang. 2026. Lite3R: A Model-Agnostic Framework for Efficient Feed-Forward 3D Reconstruction. *arXiv preprint arXiv:2605.11354* (2026). <https://arxiv.org/abs/2605.11354>
- [204] Jiahui Zhang, Yuelei Li, Anpei Chen, Muyu Xu, Kunhao Liu, Jianyuan Wang, Xiao-Xiao Long, Hanxue Liang, Zexiang Xu, Hao Su, Christian Theobalt, Christian Rupprecht, Andrea Vedaldi, Kaichen Zhou, Hanspeter Pfister, Paul Pu Liang, Shijian Lu, and Fangneng Zhan. 2026. Advances in Feed-Forward 3D Reconstruction and View Synthesis: A Survey. *Computer Graphics Forum (CGF)* (2026). <https://arxiv.org/abs/2507.14501>
- [205] Jianjun Zhang, Jian Zhu, Taiyi Su, Chong Ma, Zitai Huang, Yi Xu, and Hanli Wang. 2026. Learning 4D Geometric Priors for Inference-Efficient World Action Models. *arXiv preprint arXiv:2607.05468* (2026). <https://arxiv.org/abs/2607.05468>
- [206] Tianjun Zhang, Fengyi Zhang, Tianchen Deng, Lin Zhang, and Hesheng Wang. 2026. Keep It CALM: Toward Calibration-Free Kilometer-Level SLAM with Visual Geometry Foundation Models via an Assistant Eye. *arXiv preprint arXiv:2604.14795* (2026). <https://arxiv.org/abs/2604.14795>
- [207] Wei Zhang, Songhua Li, Yihang Wu, Qiang Li, and Qi Wang. 2026. VGGT-CD: Training-Free Robust Registration for 3D Change Detection. *arXiv preprint arXiv:2605.16859* (2026). <https://arxiv.org/abs/2605.16859>
- [208] Wei Zhang, Yihang Wu, Songhua Li, Wenjie Ma, Xin Ma, Qiang Li, and Qi Wang. 2025. Review of Feed-forward 3D Reconstruction: From DUST3R to VGGT. *arXiv preprint arXiv:2507.08448* (2025). <https://arxiv.org/abs/2507.08448>
- [209] Yipu Zhang, Jintao Cheng, Weilin Feng, Jiehao Luo, Chuanguang Yang, Zhulin An, Yongjun Xu, and Wei Zhang. 2026. Not All Tasks Quantize Equally: Fisher-Guided Quantization for Visual Geometry Transformer. *arXiv preprint arXiv:2605.15828* (2026). <https://arxiv.org/abs/2605.15828>
- [210] Yipu Zhang, Jintao Cheng, Xingyu Liu, Zeyu Li, Carol Jingyi Li, Jin Wu, Lin Jiang, Ceyu Xu, Yuan Xie, Jiang Xu, and Wei Zhang. 2026. VersaQ-3D: Architecture Support for Visual Geometry Grounded Transformers via Versatile Quantization. *arXiv preprint arXiv:2601.20317* (2026). <https://arxiv.org/abs/2601.20317>
- [211] Yuhao Zhang, Wanxi Dong, Yue Shi, Yi Liang, Jingnan Gao, Qiaochu Yang, Yaxing Lyu, Zhixuan Liang, Yibin Liu, Congsheng Xu, Xianda Guo, Wei Sui, Yaohui Jin, Xiaokang Yang, Yanyan Xu, and Yao Mu. 2026. R3DP: Real-Time 3D-Aware Policy for Embodied Manipulation. In *European Conference on Computer Vision (ECCV)*. <https://dazazh.github.io/r3dp-project-page/>
- [212] Ruosen Zhao, Zhikang Zhang, Jialei Xu, Jiahao Chang, Dong Chen, Lingyun Li, Weijian Sun, and Zizhuang Wei. 2026. SpaceMind: Camera-Guided Modality Fusion for Spatial Reasoning in Vision-Language Models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2511.23075>
- [213] Zesen Zhao, Minkyung Cho, Hui Shen, Boyuan Zheng, Kunxiao Gao, Yulong Cao, and Z. Morley Mao. 2026. Test-Time Scaling for World Action Models via Zero-Shot Geometric Evaluation. *arXiv preprint arXiv:2607.17454* (2026). <https://arxiv.org/abs/2607.17454>
- [214] Zelin Zhao, Xinyu Gong, Bangya Liu, Ziyang Song, Jun Zhang, Suhui Wu, Yongxin Chen, and Hao Zhang. 2026. CETCAM: Camera-Controllable Video Generation via Consistent and Extensible Tokenization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2512.19020>
- [215] Shuhong Zheng, Michael Oechsle, Erik Sandström, Marie-Julie Rakotosaona, Federico Tombari, and Igor Gilitschenski. 2026. Good Token Hunting: A Hitchhiker’s Guide to Token Selection for Visual Geometry Transformers. *arXiv preprint arXiv:2605.23892* (2026). <https://arxiv.org/abs/2605.23892>

- [216] Yang Zheng, Adam W. Harley, Bokui Shen, Gordon Wetzstein, and Leonidas J. Guibas. 2023. PointOdyssey: A Large-Scale Synthetic Dataset for Long-Term Point Tracking. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 19855–19865. https://openaccess.thecvf.com/content/ICCV2023/html/Zheng_PointOdyssey_A_Large-Scale_Synthetic_Dataset_for_Long-Term_Point_Tracking_ICCV_2023_paper.html
- [217] Changqing Zhou, Yueru Luo, and Changhao Chen. 2026. Generalizing Visual Geometry Priors to Sparse Gaussian Occupancy Prediction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/2602.21552>
- [218] Jingke Zhou, Xianliang Huang, Yixin Ren, Dikai Fan, Shengyu Gu, Bin Tian, Xiao Liu, and Tianjia Shao. 2026. OV3DSeg-VGGT: Open-Vocabulary 3D Segmentation with Visual Geometry-Grounded Transformers. *Visual Informatics* 10 (2026), 100311. doi:10.1016/j.visinf.2026.100311
- [219] Kaichen Zhou, Yuzhen Chen, Fangneng Zhan, Hang Hua, Grace Chen, Xinhai Chang, Ao Qu, Yilun Du, Zhuang Liu, Paul Pu Liang, and Mengyu Wang. 2026. GEM-4D: Geometry-Enhanced Video World Models for Robot Manipulation. *arXiv preprint arXiv:2605.22882* (2026). <https://arxiv.org/abs/2605.22882>
- [220] Kaichen Zhou, Yuhan Wang, Grace Chen, Xinhai Chang, Gaspard Beaudouin, Fangneng Zhan, Paul Pu Liang, and Mengyu Wang. 2026. PAGE-4D: Disentangled Pose and Geometry Estimation for VGGT-4D Perception. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2510.17568>
- [221] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyfe, and Noah Snavely. 2018. Stereo Magnification: Learning View Synthesis Using Multiplane Images. *ACM Transactions on Graphics (TOG)* (2018). doi:10.1145/3197517.3201323
- [222] Yang Zhou, Yifan Wang, Jianjun Zhou, Wenzheng Chang, Haoyu Guo, Zizun Li, Kaijing Ma, Xinyue Li, Yating Wang, Haoyi Zhu, Mingyu Liu, Dingning Liu, Jiange Yang, Zhoujie Fu, Junyi Chen, Chunhua Shen, Jiangmiao Pang, Kaipeng Zhang, and Tong He. 2026. OmniWorld: A Multi-Domain and Multi-Modal Dataset for 4D World Modeling. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2509.12201>
- [223] Alex Zihao Zhu, Dinesh Thakur, Tolga Özaslan, Bernd Pfrommer, Vijay Kumar, and Kostas Daniilidis. 2018. The Multivehicle Stereo Event Camera Dataset: An Event Camera Dataset for 3D Perception. *IEEE Robotics and Automation Letters (RA-L)* (2018). doi:10.1109/lra.2018.2800793
- [224] Dong Zhuo, Wenzhao Zheng, Jiahe Guo, Yuqi Wu, Jie Zhou, and Jiwen Lu. 2026. Streaming 4D Visual Geometry Transformer. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2507.11539>
- [225] Zichen Zou, Xiaosong Jia, Zuxuan Wu, and Yu-Gang Jiang. 2026. Attention Itself Could Retrieve. RetrieveVGGT: Training-Free Long Context Streaming 3D Reconstruction via Query-Key Similarity Retrieval. *arXiv preprint arXiv:2605.09644* (2026). <https://arxiv.org/abs/2605.09644>